

SOVEREIGN BOND RESTRUCTURING: COMMITMENT VS. FLEXIBILITY*

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Abstract

Policies facilitating sovereign bond restructuring are said to help countries emerge from distress and even reduce negative spillovers across classes of bonds. We use natural experiments to assess such policies. Our findings suggest they can backfire: Committing *not* to restructure one class of bonds is valuable not only for that class, but for others too. We develop a model to rationalize these positive spillovers. It points to a system of cross-bond equations that, we show, can be exploited to estimate unobservable elasticities in terms of a few sufficient statistics. We use this method to quantify how much commitment decreases default costs.

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1 Introduction

In early 2023, more than half of all low-income countries were in debt distress.¹ The last few years' defaulters include Argentina, Belize, Ecuador, Ethiopia, Lebanon, Mozambique, Russia, Ukraine, Sri Lanka, Suriname, and Zambia.² Others are pleading for debt relief.³ Some defaults are hard defaults in which bondholders get little or nothing. But the vast majority are soft/partial defaults in which they take a haircut but still get something in a debt restructuring.

The policy world has demanded action. The World Bank and the IMF advocate making restructuring easier, focusing on “resolving the impasse in debt restructurings” and “working hard...to avoid a disorderly debt restructuring.”⁴ The G20 is also on board, having agreed to the Common Framework for Debt Treatments, which facilitates sovereign debt restructuring by, inter alia, helping dispersed bondholders coordinate.⁵ Support for such policies is not universal. Detractors argue that they encourage opportunistic restructuring, decreasing the value of bonds and even choking off credit ex ante.⁶

Neither position, on its own, provides a full justification for outside intervention. Sovereigns already manage this tradeoff contractually, preserving some flexibility to avoid hard default while making some commitment not to restructure opportunistically. The market's widespread adoption of collective action clauses (CACs), which ostensibly benefit sovereigns and creditors alike, is a case in point (Carletti et al., 2020).

Yet, as Bolton and Jeanne (2007, 2009) point out, privately optimal contractual devices need not be socially optimal. Including them in one bond can have externalities on others. Indeed, they make an analogy between making one bond hard to restructure and installing a burglar alarm (2007, p. 903): It could provide protection for an individual only by diverting

¹See, e.g., “Why a Plan to Help Low-income Nations Rework their Debt Isn’t Working” (*Bloomberg*, Feb 14, 2023).

²See, e.g., “Argentina clinches near-unanimous backing for debt restructuring” (*Financial Times*, August 31, 2020), “How to restructure sovereign debt” (*Financial Times*, September 16, 2022), and “Ethiopia defaults on sovereign debt after deadline expires on \$33 mn payment” (*Financial Times*, December 27, 2023).

³“Emerging economies plead for more ambitious debt relief programmes” (*Financial Times*, October 11, 2020).

⁴See blogs.worldbank.org/voices/action-needed-debt-crisis and imf.org/en/News/Articles/-current-sovereign-debt-challenges-and-priorities-in-the-period-ahead.

⁵See imf.org/en/About/FAQ/sovereign-debt; see also Krueger (2002) and White (2002) for earlier proposals.

⁶See, e.g., Dooley (2000) and Shleifer (2003).

costs onto others—other homes end up being targeted by burglars—making it self-defeating collectively.

The policy debate needs to know how the commitment not to restructure one bond affects the value of all of a sovereign’s debt. (i) Is it valuable for that bond? (ii) Does it have externalities on other classes? The answers remain elusive. The evidence on the first question is mixed, perhaps due to inherent identification challenges—bonds made hard to restructure (e.g., via foreign law or cross-default clauses) could differ in any number of ways from those made easy (e.g., currency, maturity, and other covenants).⁷ There is no evidence on the second question as yet (to our knowledge).

In this paper, we bring evidence from natural experiments to bear on these questions. Our results suggest that the commitment not to restructure one bond is valuable not only (i) for that bond but (ii) for others as well: Commitment has positive cross-bond spillovers—it seems more like a successful neighborhood crime watch, providing collective protection against restructuring, than a self-defeating burglar alarm, deflecting it onto others.

We develop a model extending Bolton and Jeanne (2007). Our model can both explain our findings and square them with the results in the literature, notably that bonds that are ostensibly easier to restructure can trade at a discount (Chamon et al., 2018), consistent with Bolton and Jeanne (2007), or at a premium (Carletti et al., 2020). True, sometimes making one bond harder to restructure can lead a sovereign to target another instead; that is the burglar alarm. However, we show it can also discourage the sovereign from restructuring any bonds at all; that is the crime watch.

The model points to cross-bond relationships that, we show, can be exploited to quantify the impact of experiments and to estimate otherwise unobservable elasticities in terms of a few sufficient statistics. Although the model is stylized, applying the methodology delivers sensible numbers, which could be useful for quantitative (i.e. less stylized) models of sovereign debt. It can also be useful for counterfactual policy analysis, as we illustrate by applying it to estimate

⁷Chamon, Schumacher, and Trebesch (2018) and Clare and Schmidlin (2014) consider heterogeneity in governing law; Becker, Richards, and Thaicharoen (2003), Bradley and Gulati (2014), Carletti et al. (2020), Eichengreen and Mody (2004), and Picarelli, Erce, and Jiang (2019) consider heterogeneity in covenants.

how much an increase in commitment decreases default costs.

Our starting point is the collective action problem inherent in bond restructuring. As we summarize in Section 2.1, dispersed bondholders could reject a restructuring that would make them better off collectively (the “hold-out problem”) or accept one that would make them worse off (the “hold-in problem”). Debtors mitigate/exploit the problems using effective threats and bribes of uncertain legality to pit bondholders against one another.

Our baseline experiments are court rulings determining whether or not such restructuring devices are legal. The first experiment, which is new to the economics literature (to our knowledge), is a case heard in the English High Court against a firm that had cut every coupon that came due for two years using a device called “consent payments,” an effective bribe in the standard restructuring toolkit that had never been tested in court. That changed when one bondholder held out and sued. The judge ruled against him in a “landmark ruling,” ratifying market practice, weakening prospective creditor bargaining power, and thereby making all outstanding bonds under English law—including sovereign bonds—easier to restructure, at least at the margin.⁸

The second experiment, which is taken from Hébert and Schreger (2017), is a sequence of rulings culminating in a case heard in the Southern District of New York about Argentina’s threat to punish hold-out bondholders, paying them nothing while servicing bonds that agreed to a haircut. The judge deemed the threat illegal and offered bondholders novel ways to enforce payment. Thus the ruling, strengthening creditor bargaining power, made all outstanding bonds issued under NY law harder to restructure and precipitated another Argentine default. (We drop Argentine bonds everywhere as, unlike Hébert and Schreger (2017), we use the experiment as a shock to NY law, not to Argentina.)

These experiments cover both foreign laws that sovereigns commonly issue under, both bribe- and threat-like devices, and both increases and decreases to commitment. They provide a laboratory to address our questions on sovereign debt restructuring. Using them, we can compare the same bonds on the trading days before and after the rulings and thereby avoid

⁸See “The consent of the (bondholder) governed” (*Financial Times*, April 22, 2013) and White & Case (2013).

relying on comparisons of different bonds. Both court decisions mattered for hundreds of sovereign bonds, as many countries issue under English or NY law. Other countries do not. Their bonds provide us with a control group for difference-in-differences (DiD) analyses.

We use a different treatment group to address each of our two motivating questions, on how making one class of bonds harder to restructure affects (i) that class—the “direct effect” hereinafter—and (ii) other classes—the “indirect/spillover effect.” For the direct effect, we consider “directly treated” bonds, defined as those issued under the affected law (English or NY). For the spillover effect, we consider “indirectly treated” bonds, defined as other bonds issued by countries with some outstanding directly treated ones.

We then compare the differences in credit spreads of each treatment group to those of the control group before and after the rulings. For the English-law experiment, we find that the direct effect of an increase in commitment is about 5 bps and the indirect effect is about the same size. For the NY-law experiment, which includes multiple events, we aggregate them by weighting them by the (signed) change in the risk-neutral default probability P^{AR} , as Hébert and Schreger do. We find that for every 1 percentage point increase in P^{AR} , the direct effect is about 0.34 bps and the indirect effect is about 0.16.

While these effects are large enough to be statistically and economically significant, they reflect only marginal changes in the law, not an overhaul of market practice. That proves useful below, where we use a linear approximation around the equilibrium of our model that might not be valid for a large change (that exercise suggests our effects are appropriately small-ish but not negligible). Besides, marginal shocks are ideal for understanding policy interventions given the prevailing equilibrium; larger shocks, in contrast, can induce SUTVA violations.

We interpret our results as causal effects of the rulings on treated bonds. To do so, we rely on two assumptions. The first is that the experiments do not affect control bonds. Our research design ensures that it holds by focusing on experiments specific to restructuring English- or NY-law bonds and choosing control bonds issued by countries without any treated bonds. The second assumption is that, conditional on control variables, treated and control bond outcomes would have moved the same way absent the experiments. Our research design ensures that

it holds by including region-, currency-, and maturity-time fixed effects as control variables, focusing on narrow (two-day) event windows, and matching treated and control bonds on risk and other characteristics. Thus, to threaten our identification, a confounding event must be nearly simultaneous with our experiments and must affect treatment and matched control bonds differently within the same region, currency, and maturity. That sounds unlikely *prima facie*. But nonetheless we do a data-driven placebo-type test to confirm as much; our results almost never arise spuriously in our sample (see Section 3.4).

Our interpretation also relies on the assumption of market efficiency, namely that short-term price changes reflect new information, not merely transient inefficiencies due to, e.g., overreaction or fire sales. The assumption is relatively weak for the bond markets we study, which are particularly informative empirically (e.g., Philippon, 2009), partly because they are frequented by expert investors; these investors get realtime updates from reporters in court via services like Debtwire and Reorg or contacts and newsletters from law firms. Moreover, the markets have been attuned to changes in governing law, like those in our experiments, since at least the Greek restructuring, when newspapers said things like, “We also really, really want to know if the new bonds under this new offer will be governed under English law or will remain under Greek law.”⁹ Nonetheless, where possible, we show that our results do not rely on the assumption of market efficiency: We repeat our analysis with a longer event window (four trading days). Our results are robust.

Despite the breadth of our baseline experiments (see above), we provide further assurance of external validity, repeating our analysis using a similar (though more imperfect) experiment, on yet another type of restructuring device: exit consents (Section 5). We find broadly analogous results.

Next, we develop a model of sovereign restructuring based on Bolton and Jeanne’s (2007), which serves to investigate the mechanism behind our results, to interpret their magnitudes, and to understand policy counterfactuals. In the model, the sovereign has two types of debt, which we call “rigid” and “flexible,” and can default on either type strategically, destroying a

⁹“Who Wants to Be a Greek Bond Holdout” (*Financial Times*, October 26, 2011).

fraction of output and inducing bargaining with creditors. The model nests Bolton and Jeanne (2007) and Carletti et al. (2020) as special cases, *mutatis mutandis*.

Our analysis reveals that there are two regimes, depending on parameters, one that lines up with the theoretical literature and another that explains our empirical findings as well as other existing ones. There is a selective default regime, in which making rigid debt harder to restructure deflects default onto flexible debt, per Bolton and Jeanne’s burglar alarm. But we show that there is another regime in our model, which we call “cross-default.” In this regime, the sovereign optimally defaults on all debt or none. In this regime, making rigid debt harder to restructure can make default unattractive altogether, increasing the likelihood that flexible debt is repaid. That is what we call the “crime watch,” which is new to this paper.

In our experiments, we interpret rigid bonds as foreign-law, flexible as local, and, per our analysis in Section 2.1, our experiments as changes in the expected recovery of rigid bonds arising from shifts in the bargaining power of rigid creditors.¹⁰ The spillovers we find empirically suggest that the crime watch, and thus the cross-default regime, could be relevant for practice and policy despite being all but omitted from canonical models.

The model can also help us to interpret our experiments, which do not immediately lend themselves to quantification—some rulings made restructuring easier, others harder, but by how much did they affect the expected haircut h_r ? We use the model to back out this number. We derive a formula for the implied change in h_r in terms of our estimated yield changes and a few other sufficient statistics, notably estimates of risk-neutral default probabilities (from CDS) and expected haircuts. We estimate that average haircuts moved by around 33 bps from the High Court ruling and by around 25 bps for each event in the Argentine saga on average. These estimates accord with our intuition that the rulings are meaningful, as they reflect landmark legal decisions, but their immediate economic effects are not large, as they pertain to specific restructuring devices.

We also show how to use the same sufficient statistics to translate our baseline estimates

¹⁰Local law bonds are deemed easier to restructure because, *inter alia*, sovereigns can change the law as Greece did in 2012 (Zettelmeyer, Trebesch, and Gulati (2014)).

into elasticities that could be useful for policy counterfactuals: We estimate that a 1% increase in haircuts increases the (risk-neutral) default probability by 17% for English law and 13% for NY (or 3.0 and 1.4 percentage points). A back-of-the-envelope application of this number, in combination with Hébert and Schreger’s (2017) estimates of the costs of an Argentine default, suggests that a one percentage point decrease in haircuts on Argentine debt would save around 5 billion USD in deadweight costs.

Our crime watch spillover, which is the result of cross-default, may appear at odds with selective default, a much-discussed phenomenon in sovereign debt markets, at least historically (Erce and Mallucci, 2018). But it should not. Cross-default in our model constitutes an attempted restructuring on multiple classes of bonds. That could still register as a selective default empirically if one class manages to hold out. Moreover, many selective defaults are closet cross-defaults, in which a sovereign imposes a haircut explicitly on foreign-currency debt but only implicitly on local-currency debt, via devaluation (see Du and Schreger (2016), Engel and Park (2022), Reinhart and Rogoff (2011), and Sunder-Plassmann (2020)).

Finally, a proviso: Our results are about the benefits of the commitment not to restructure for creditors, not sovereign borrowers. It could be good for them, e.g., facilitating access to credit, or bad, e.g., limiting the self-insurance role of default (Zame, 1993). Direct evidence on the outcomes for sovereigns requires further research. For example, you could look at what sovereigns do following a shock to creditor commitment, something our experiments are unfortunately not well suited to do: Their strength in identifying price responses—a narrow event window—is a weakness in identifying behavioral responses, which do not happen fast enough for us to pick up. Nonetheless, our findings that hard-to-restructure debt is cheaper for sovereigns and can even make other debt cheaper resonates with observed behavior. Indeed, sovereigns choose to issue hard-to-restructure debt and even, at times, try to engineer spillovers.¹¹

¹¹E.g., in 2012, Greece seemed to try to make its private debt harder to restructure by tying it to its multilateral debt (Zettelmeyer, Trebesch, and Gulati, 2014); cf. Schlegl, Trebesch, and Wright (2019).

2 Institutional Background and Data

Here we lay out the ingredients behind our analysis, starting with a primer on the practice and problems of debt restructurings, including a description of the threats and bribes used to impose haircuts on creditors and the legal remedies creditors use to resist them (Section 2.1). We also derive a relationship between equilibrium haircuts and the strength of such remedies that is useful to map our theory to our experiments, which are summarized in the next section (Section 2.2). Finally, we describe the data we use (Section 2.3).

2.1 Threats and Bribes in Bond Restructuring

Sovereign default is not uncommon. As Reinhart and Rogoff (2009) catalog, “[m]ost countries in all regions have gone through a prolonged phase as serial defaulters” (p. 49). Most defaults are partial, with creditors taking haircuts but still getting something (Arellano, Mateos-Planas, and Ríos-Rull (2019), Meyer, Reinhart, and Trebesch (2022), and Cruces and Trebesch (2013)).

Given bondholders are often dispersed, bond restructurings are plagued by collective action problems. On the one hand, there is what is known as the “hold-out problem”: An individual bondholder could reject an offer that could make bondholders collectively better off. On the other hand, there is what is sometimes called the “hold-in problem”: An individual bondholder could accept an offer that makes them collectively worse off.¹² (See Buchheit and Gulati (2000), Donaldson et al. (2022), Gertner and Scharfstein (1991), Kahan and Tuckman (1993), and Roe (1987).)

To see how the hold-out problem works, consider a sovereign that has so much debt that it is tempted to default outright. Collectively, bondholders might be better off taking a haircut to reduce the sovereign’s debt burden and thereby increase the likelihood it repays them at all. But an individual bondholder still might not accept the haircut. By holding out, it can free ride on others’ accepting, benefiting from the overall debt reduction without taking a haircut itself. As all bondholders have an incentive to act this way, the whole restructuring can fall

¹²These are not the only ways that collective action problems can lead to inefficiencies in sovereign-debt restructuring, something Pitchford and Wright (2012) demonstrate in a model of bargaining delays.

apart even though it would have made everyone better off.

Sovereigns tackle the hold-out problem with threats and bribes. One threat is to pay nothing to “hold-out” bondholders that do not accept a haircut, as Argentina did in 2001 (see Section 2.2). But the legality of such threats is questionable, and the hold-outs might litigate and end up getting paid in full. In equations, if a bondholder with face value D accepts, its face value is decreased by the haircut h . If it holds out, it risks getting paid nothing, but could litigate and get paid in full with probability π . Thus it accepts if

$$(1 - h)D \geq \pi D, \tag{1}$$

or $h \leq 1 - \pi$. Haircuts are higher when hold-outs are less likely to win in court. Conversely, precedent rulings in favor of creditors raise their bargaining power in a restructuring, a relationship we employ off the shelf in the model below (see Section 4).

Such threats can be especially effective if invoked using a collective action clause (CAC), a contractual term, now commonplace in sovereign bonds, that addresses free rider problems by allowing all bonds in a series to be modified by a specified supermajority, so minority hold-outs effectively cannot litigate (π is close to zero in equation (1)).¹³ But, while helpful (Fang, Schumacher, and Trebesch (2021)), CACs are not a panacea. Free rider problems persist when CACs cannot be aggregated across series (Gelpern, Heller, and Setser (2016)), and they “might be an invitation to litigation” otherwise.¹⁴

Bribes are the alternative to threats. One contractual device, so-called “consent payments,” constitutes a payment to bondholders when they agree (“consent”) to changes in terms (of all bonds in a series). However, legality is again uncertain and hold-outs could likewise litigate for full payment.¹⁵ In equations, if a bondholder with face value D accepts, it gets the (propor-

¹³Most countries enshrine some bondholder rights in legal statutes for corporate bonds (e.g., the Trust Indenture Act in the US). Not so for sovereign bonds, in which they are specified, by contract, in the indenture; see example prospectuses: under NY, English, and Italian laws, respectively: [Suriname](#), [Poland](#), and [Italy](#).

¹⁴creditslips.org/creditslips/2020/06/the-argentine-re-designation-drama-notes-from-two-frustrated-readers.

¹⁵Panizza, Sturzenegger, and Zettelmeyer (2009) enumerate cases of such litigation; hold-outs received full payment or close to it in some cases but little or nothing in others. Mitchener and Trebesch (2023) show that threats of litigation are a prominent feature of recent sovereign distress episodes. See Schumacher, Trebesch, and Enderlein (2021) for

tional) bribe b but decreases its face value by a haircut h . If it holds out, it forgoes the bribe but can litigate and be paid in full with probability π . Thus it accepts if

$$bD + (1 - h)D \geq \pi D + (1 - \pi)(1 - h)D, \quad (2)$$

or $h \leq b/\pi$. The takeaway of the analysis of bribes is the same as of threats: Haircuts are higher when hold-outs are less likely to win in court, so we can apply the relationship in the model later regardless of the reason for litigation.

Overall, such threats and bribes are a double-edged sword. By making it easier to restructure, they mitigate the hold-out problem. But they can exacerbate its counterpart, the hold-in problem (especially when combined with other anti-hold-out provisions like CACs), because they can force bondholders into a prisoner’s dilemma in which they tender even if restructuring is against their collective interest. Either way, what haircuts are feasible depends on how likely a legal jurisdiction is to protect bondholders—on how big π is.

2.2 Baseline Rulings: *Azevedo v Imcopa* and *Argentina v NML Capital*

At the center of our analysis are two legal disputes over bond restructurings, one under English law about bribes, and another under NY law about threats.

2.2.1 *Azevedo v Imcopa: Bribes under English Law*

Our first baseline experiment is the May 2012 ruling by the English High Court upholding the legality of consent payments—an effective bribe that has proved useful in sovereign debt restructuring.¹⁶ The ruling thus made it easier to restructure bonds in exchange offers.¹⁷ It applied to all English-law bonds, including foreign sovereign bonds, but was otherwise unrelated

an overview of sovereign-debt litigation in recent decades.

¹⁶E.g., Suriname deployed it repeatedly in 2020 and 2021 to defer interest payments on two bond series; see reporting by *Fitch* ([fitchratings.com/research/sovereigns/fitch-upgrades-suriname-lt-fc-idr-to-cc-16-07-2020](https://www.fitchratings.com/research/sovereigns/fitch-upgrades-suriname-lt-fc-idr-to-cc-16-07-2020)) and *LatinFinance* ([latinfinance.com/daily-briefs/2020/11/17/suriname-seeks-breathing-room-for-debt-payments](https://www.latinfinance.com/daily-briefs/2020/11/17/suriname-seeks-breathing-room-for-debt-payments)). Primary sources are available at [bourse.lu/security/US86886PAB85/299502](https://www.bourse.lu/security/US86886PAB85/299502).

¹⁷See *Azevedo v Imcopa Importacao, Exportacao e Industria de Oleos Ltda* [2012] EWHC 1849 (Comm) (30 May 2012)).

to the sovereign bond market.

The case, brought by two individual bondholders, Sergio Barreiros Azevedo and Vera Cintia Alvarez, against Imcopa Group, a Brazilian soybean company, constitutes a “landmark decision,” according to the *Financial Times*.¹⁸ The newspaper also stresses that consent payments, while not uncommon, had never been considered by English courts, making the case “hugely important.” Despite attention in the law literature on both corporate (e.g., Liu (2017)) and sovereign debt (e.g., Drake (2014) and Kupelyants (2018)), it is new to the finance and economics literature. Hence, we offer a précis now.

In 2008, soybean prices plummeted from over \$16 a bushel in July to under \$8 in December.¹⁹ Imcopa embarked on a plan to reduce its interest payments by restructuring its English-law bonds. Over the next two years, it restructured its bonds every time an interest payment came due. Each time, it offered bondholders consent payments if they agreed to put off the interest payment. As predicted by the prisoner’s dilemma described above (Section 2.1), nearly all bondholders accepted.

But the last time, Azevedo and Alvarez held out. Per equation (2), they found the bribe b insufficient compensation for the chance π of full repayment via litigation. They sued Imcopa in England, claiming inter alia that the consent payments constituted an illegal bribe.

The High Court ruled for the defendant on May 30, 2012: Since consent payments were offered to all bondholders, they were not an illegal bribe. The decision ratified current practice rather than upending it, so it did not shake up the bond market—giving us a clean experiment free of confounding general equilibrium reactions. It was not a foregone conclusion, however: Just two months later, the High Court ruled the other way in the analogous case of *Assenagon Asset Management v Irish Bank Resolution Corp* (cf. Section 5.1).

¹⁸See supra note 8.

¹⁹See, e.g., macrotrends.net.

2.2.2 Argentina v NML Capital: Threats under NY Law

Our second baseline experiment is a series of court decisions throughout the Argentine restructuring saga, culminating in a 2014 Manhattan federal district court ruling that made it harder to punish hold-out bondholders.

In 2001, Argentina missed payments on NY-law bonds with face value of roughly \$82bn. It then offered bondholders a choice: Accept a 70% haircut or hold out and receive nothing. Per equation (1), over 90% accepted. Argentina made good on its threat, servicing only its restructured debt.

Litigation ensued. Most notably, hedge fund NML Capital sued Argentina in New York for full repayment, arguing that servicing restructured debt while defaulting on hold-outs violated the *pari passu* clause in the bond indenture. The presiding judge, Thomas P. Griesa, ruled for the plaintiff, blocking Argentina from servicing any debt until hold-out bonds were paid in full. Argentina appealed, counter-sued, and attempted to service its debt outside US jurisdiction, but the US judiciary affirmed and blocked all workarounds. Argentina defaulted on bonds serviced via US payment systems in July 2014 and resumed payments only in April 2016, after settling with NML Capital and other hold-outs.

Following Hébert and Schreger (2017), we use fifteen non-overlapping, unconfounded events from this legal battle. We sign each event’s direction—debtor- or creditor-friendly—using changes in the risk-neutral probability of default on Argentine bonds from Hébert and Schreger (2017). Eight decisions go against Argentina (default probability rises) and seven against creditors. The most consequential creditor-friendly event is the Supreme Court’s denial of Argentina’s appeal on June 16, 2014, when the default probability jumped 7.72 percentage points (and the stock market fell 6.50%). The decision barred US intermediaries from distributing payments to bondholders on Argentina’s behalf. Two weeks later, Argentina missed a coupon payment. The most consequential debtor-friendly event is the Appeals Court’s emergency stay of Judge Griesa’s order on November 28, 2012, when the default probability dropped 10.61 percentage points (and the stock market rallied by 6.75%). The average absolute change in

default probability across all fifteen events is 3.93%. A full list and description appears in Table A19 of the Internet Appendix to Hébert and Schreger (2017).

2.3 Data Sources and Sample Construction

Bond data come from Dealogic DCM and Bloomberg’s records of sovereign bond issues from 1980 onwards. We collect data on bonds outstanding between May 2012 and October 2014, a sample period that includes both baseline experiments, as well as two others used for external validity (Section 5). The sample starts after the Greek restructuring in early 2012, an event which made it salient that sovereign debt haircuts can depend on governing law.

We focus on bonds issued by countries with significant English- and NY-law borrowing and those issued by relevant control countries, e.g., countries in Europe and Latin America. We include only bonds with some data available on governing law, our key variable of interest. Dealogic provides it for most issues; Bloomberg for fewer.

We must make a few adjustments when we merge the governing law data from the two sources, as sometimes multiple laws are listed. Occasionally, (i) Bloomberg lists two laws. In this case, if Dealogic provides a single governing law for the bond, we use that law; if not, we use the one that coincides with the currency denomination of the bond. Other times, (ii) Bloomberg and Dealogic list different laws. In this case, if one coincides with the currency denomination, we use that; if not, we use the one provided by Dealogic. (See also Kropp, Gulati, and Weidemaier (2018) for a discussion of data sources and quality for governing law.)

We exclude bonds with missing ISIN and those issued by publicly-owned companies, state/local governments, or Argentina, which, as of 2012, was still in litigation following its 2001 default and subsequent restructuring. The sample of bonds with available governing law and credit spread information that are outstanding during the sample period contains 2,024 bonds issued by 76 countries, denominated in 23 currencies, and 907,415 bond-day observations.

We collect daily yields and swap rates from Bloomberg, which we use to construct credit spreads and thereby difference out risk-free rate changes.

For our English-law experiment, our event window is two trading days: from May 29 to May

31, 2012. Since bonds are traded OTC in different time zones, this is the narrowest window that ensures that bonds traded in European, Middle Eastern, and Asian markets reflect the new information. (We also consider a wider event window as a robustness check to show our results are not an artifact of short-term market overreaction; see 3.4.) On May 31, the sample contains 1,281 bonds, issued by 67 countries in 22 currencies. We winsorize credit spread changes at 1% and 99%.

For our NY-law experiment, we supplement these data with data on changes in the estimated risk-neutral probability of Argentine default on fifteen event-days throughout the saga—including not only NML’s filing and the court’s ruling, but also Argentina’s appeals and counter-suits. We take these from Hébert and Schreger (2017), along with the associated two-day event windows. The first event window ends on November 23, 2012. On that day, the sample contains 1,325 bonds, issued by 67 countries in 23 currencies. Figure 1 plots treated and control countries.

Table 1 shows the summary statistics, aggregated by treatment group, for bonds outstanding on the day before (the start of) each experiment. The median number of bond series a country has outstanding is 12 in May 2012 and 14 in November 2012; the median face value is around \$1bn on both dates; the median maturity is about 8 years on both dates; the median credit spread is 2.87% in May and 2.11% in November. The average amount of debt issued under English law is about a third, under NY-law about a quarter, and under local about 39%. Table SA.6 in the Supplementary Appendix shows statistics by country on the date of the English-law experiment.

3 Empirical Framework and Results

Here we set up our empirical specifications (Section 3.1), we argue they have a causal interpretation (Section 3.2), we report our baseline empirical results (Section 3.3), and we argue for their robustness (Section 3.4).

3.1 Estimation Strategy

To evaluate the bond market response to a change in the ease of bond restructuring, we use a high-frequency difference-in-differences (DiD) approach. Our baseline model, which we label (R1), nests two DiD specifications, each addressing one of our motivating questions.

In each of the two DiD specifications nested in regression (R1), the first difference is an increase in commitment not to restructure over an event window, which we label ΔComm_t . The second difference is between our control bonds and one of two treatment groups. The first is “directly treated” bonds, namely those issued under English and NY law, for each respective experiment. The second is “indirectly treated” bonds, which we define as other bonds issued by countries with *any* outstanding directly treated bonds, a definition that avoids issues with continuous treatment, such as sensitivity to activist minority bondholders and other non-monotonicities.²⁰ Nesting these specifications in a single regression has the advantage of estimating the coefficients on control variables on the full sample (absent controls the nested specification is equivalent to two separate DiD specifications). Formally, we estimate:

$$\begin{aligned} s_{i,t} = & \beta_1 \Delta\text{Comm}_t \times \mathbb{1}_i^{\text{Direct}} + \beta_2 \Delta\text{Comm}_t \times \mathbb{1}_{i,c}^{\text{Indirect}} \\ & + \beta_3 \mathbb{1}_i^{\text{Direct}} + \beta_4 \mathbb{1}_{i,c}^{\text{Indirect}} + \alpha^\top \text{FE}_{i,t} + \varepsilon_{i,t}. \end{aligned} \tag{R1}$$

In our baseline specifications, the dependent variable s_{it} is the change in the credit spread. Relative to other measures of default risk (viz. yields), the credit spread has the advantage of isolating risk by differencing out the risk-free rate. We construct the credit spread by subtracting the maturity- and currency-matched swap rate from the yield, as detailed in Appendix

²⁰These types of issues would arise if, say, we used a continuous variable like the fraction of directly treated bonds a country has outstanding. A continuous treatment measure is a poor proxy if some bondholders exert influence even with small holdings. Witness the Argentine saga, when the refusal to pay hold-outs induced mass default (Section 2.2.2) and an episode in the Greek restructuring, when the decision to pay a select group of bonds was deemed “scandalous” (“In about-face, Greece pays bond swap holdouts” (*Reuters*, May 15, 2012)). We also show how such a specification can give spurious results even in theory because the desired treatment effect need not be monotonic in the proportion of foreign-law debt. The reason, which can be seen in our model, is that the measured effect with a continuous treatment—a mixed derivative—depends on the slope of the density at the default threshold—denoted by $f'(z)$ for $z \in \{z^{\emptyset \succ rf}, z^f \succ rf, z^{\emptyset \succ f}\}$ below—which can be arbitrarily large, positive or negative.

A.2.²¹ (These baseline specifications compare changes, the analog of “abnormal returns,” in and out of the event window, as is standard in event studies; however, as a robustness check, we also compare levels immediately before and after the event (see Table 5), as is more common in DiDs.)

The key independent variable, ΔComm_t , is defined as follows. For the English-law experiment, which is a single event that made restructuring easier (Section 2.2.1), the variable is just the indicator of the event with a minus sign: $\Delta\text{Comm}_t = -\mathbb{1}_t^{\text{Event}}$, where $\mathbb{1}_t^{\text{Event}} = 1$ if date t is the first trading day after the High Court ruling; for the NY-law experiment, which is a sequence of events, some making restructuring easier and others harder, we weight the events by their (signed) intensity by multiplying the indicator by the change in the estimated risk-neutral probability of default: $\Delta\text{Comm}_t = \Delta P_t^{\text{AR}} \mathbb{1}_t^{\text{Event}}$, where, analogously, $\mathbb{1}_t^{\text{Event}} = 1$ if date t is the end of one of the two-day event windows cataloged in Hébert and Schreger (2017).²² The remaining dummy variables are as follows: $\mathbb{1}_i^{\text{Direct}} = 1$ if bond i is in the direct treatment group; $\mathbb{1}_{i,c}^{\text{Indirect}} = 1$ if bond i is in our indirect treatment group. Their coefficients, estimated on the entire sample instead of just around event days, serve to de-trend the credit spreads in each group.

The terms $\text{FE}_{i,t}$ denote vectors of fixed effects, including region-, currency-, and maturity-time for panel specifications, and pair-time for matching specifications (detailed in Table 3).²³ We also control for bond risk, either by including granular credit-rating FEs or by running the specification on matched bond pairs, comparing English- to NY-law bonds for the direct effect and local-law bonds of English-law issuers to local-law bonds of non-English-law issuers for the indirect effect. We match bonds by currency and maturity. We also match on credit rating, as tightly as we can given the sample size; for the English-law experiment that is within two

²¹Swap rates avoid measurement problems arising from credit risk and convenience yields in seemingly risk-free sovereign bonds (Du and Schreger (2016); Kremens (2024)).

²²We take the time series of the change in default probability ΔP_t^{AR} directly from Hébert and Schreger (2017). See their Appendix A.4 for its construction. Our interpretation, while consistent with this methodology, does not hinge on its details. It requires only that the measure has the correct (i) sign (to indicate whether a ruling was in the creditors’ or debtor’s favor) and (ii) ranking of events (to indicate which rulings were more relevant than others).

²³The regions are Europe, South America, Central America, Middle East/North Africa, Africa, Central Asia, East Asia (see Appendix A for details); the maturity fixed effects refer to two-year maturity buckets.

notches (so, e.g., a bond rated AA− may be matched to one rated as high as AA+ or as low as A) and, for the NY-law experiment, that is within one notch for the direct effect and within the same rating (so it is equivalent to a notch-level fixed effect) for the indirect. The details of the matching procedure are in Appendix B. Table 2 shows balance tests for the matched bond pairs: Treated bonds do not systematically differ from control bonds on default risk (proxied by credit spreads), maturity, or rating, but their issuers have slightly higher debt-to-GDP ratios.

Of the two specifications with risk controls, matching has three advantages vis-à-vis ratings FEs: (i) It allows us to limit the estimation of the direct effect to foreign-law bonds, matching treated English-law bonds to NY-law control bonds and vice versa; (ii) it makes the comparisons transparent (see Table SA.4 and Table SA.5 and in the Supplementary Appendix for the list of bond pairs); and (iii) it has slightly more statistical power (cf. Section 3.3.1). Hence we favor it below.

The key coefficients of interest are β_1 and β_2 . They measure the changes in s_{it} around the event dates for, respectively, directly treated and indirectly treated bonds relative to our control groups. They represent averages across heterogeneous bonds conditional on the market conditions—countries’ debt levels, investors’ risk aversion, etc.—on the event date; i.e. they are local average treatment effects (LATE).

3.2 Identifying Assumptions

Our empirical strategy is a difference-in-differences (DiD) over a narrow event window. We can interpret the estimates as causal results of the rulings if the following assumption is satisfied: The expected change in the dependent variable over the window, conditional on our controls (but not on the ruling), must be equal for treated and control bonds. The assumption could be violated if there were another event that both (i) happened at the same time as our events and (ii) affected our treatment and control bonds differently.

The assumption, often referred to as “parallel trends,” is inherently untestable.²⁴ But we

²⁴A common approach in DiD settings is to test for parallel *pre*-trends as a proxy for counterfactual post-event realizations. This approach is ill-suited in our setting where the dependent variable, an asset price, is inherently forward looking. If, for instance, bond market participants anticipated the rulings as they drew closer, this would

can mitigate concerns about it. We do so in five principal ways.

- (a) We include region-time, currency-time, maturity-time, and risk (credit rating)-time fixed effects in our panel regressions. These absorb variation on any given date common to a given region, currency, maturity bucket, or risk category.
- (b) We include a specification run on only matched pairs of closely comparable bonds which we list entirely in the Supplementary Appendix (Table SA.4 and Table SA.5). These bonds are in the same currency and are matched on maturity and credit rating.
To be a threat to identification, confounding shocks would have to hit matched bonds in systematically different ways.
- (c) We choose a narrow event window of just two days. To be a threat to our identification, an event must both affect treatment and control bonds differently and also happen at nearly exactly the same time as the ruling. We have searched the news and have not found any such relevant event released around the rulings.
- (d) We control for movements in risk-free rates by using credit spreads as our dependent variable (subtracting currency- and maturity-matched swap rates from bond yields).
- (e) We include multiple experiments; the NY-law experiment uses different bonds, at different dates, from the English-law one, and itself includes multiple events, some increasing commitment, others decreasing it.

To threaten our identification, something must both coincide with our experiments and affect treated and control bonds differently given our controls/matching. For example, our interpretation would not be strictly justified if there were differences in liquidity of treated and control bonds not taken care of by (a) and (b) above and there were shocks to the market price of

generate a pre-trend without undermining the experiment. Instead, it would reduce the residual treatment intensity and thus bias against finding a result. However, pre-trend tests may still be a useful diagnostic if the concern is that markets are far from efficient and driven by high-frequency microstructure noise. In such markets, uninformative short-term trends could, in principle, differ systematically between treated and control bonds. As such, with the proviso that our findings should be interpreted cautiously, we test for pre-trends in the English-law experiment. We find none (Figure SA.1). (Such a test is infeasible for the NY-law experiment where some events occur in quick succession.)

liquidity that occurred at the same time as the experiments *as well*. Such concerns seem far fetched to us. But we address them in robustness checks nonetheless (Section 3.4).

For the NY-law experiment, the events did not only make NY-law bonds harder to restructure, but also made Argentina more likely to default. Thus, although the conditions above suffice to say our findings are caused by the rulings, to interpret them as a result of bonds becoming harder to restructure requires an additional assumption, albeit an apparently weak one: Countries' exposure to Argentine default is uncorrelated with their propensity to borrow under New York law within region/after matching. (Recall that we drop all Argentine bonds from our sample (Section 2.3) and note that we also drop all South American bonds in a robustness test (Section 3.4.2).)

3.3 Do Sovereign Creditors Benefit from Committing Not to Restructure?

Table 3 reports the results from estimating equation (R1), our DiD regression of spread reactions on directly and indirectly treated bonds for both the English-law and NY-law experiments.

3.3.1 *The Effect on Directly Treated Bonds*

The direct treatment effect for the English-law experiment ranges from about 5–11 bps, with larger effects for the specifications without risk controls. The change in coefficient across specifications could reflect heterogeneous treatment effects: As treated countries have both higher risk and larger effects, the specification without risk controls (which compares high- and low-risk bonds) yields a larger coefficient. The two specifications with risk controls (FEs and matching) both deliver coefficient estimates of 5–6 bps, both of which are statistically different from zero at the five percent level. But the standard errors are larger in the fixed-effects specification as the estimates sometimes rely on comparisons among relatively few bonds within a single ratings notch. That is one of the reasons we favor the matching specification hereinafter, per the discussion above (Section 3.1).

The direct treatment effect for the NY-law experiment is about 0.25–0.34 bps for every 1 percentage point increase in the risk-neutral probability of Argentine default, with coefficients

and standard errors relatively stable across specifications, irrespective of risk controls.

The results suggest increased commitment benefited bondholders. In the language of Section 2.1, that means bondholders care more about protection against the hold-in problem than the risk of the hold-out problem.

3.3.2 The Spillover Effect on the Indirectly Treated Bonds

The indirect treatment effect for the English-law experiment ranges from about 4–13 bps. The coefficients and standard errors demonstrate the same patterns as for the direct effect (and we offer the same explanation for them).

The indirect treatment effect for the NY-law experiment is about 0.12–0.16 bps for every 1 percentage point increase in the risk-neutral probability of Argentine default. Again both the coefficients and standard errors are similar across specifications.

This result suggests a positive spillover from increased commitment, in line with harder-to-restructure bonds disciplining a sovereign, helping holders not only of those bonds but of others as well. That interpretation does not require that foreign-law bonds are harder to restructure than local-, but the assumption does help explain why the indirect effect can be similar in size to the direct, a result verified in a within-country event study in Table 7, shown to be consistent with the disciplining role of rigid/foreign-law debt in Section 4.2.

3.4 Empirical Concerns and Robustness Tests

We now address residual concerns about our baseline tests with a number of robustness exercises.

3.4.1 Concern: Spurious Results (Omitted Variables and/or “Noise”)

One concern is that our results are spurious, caused not by the experiments we study, but by something else, unknown to us, that happened in the same event window(s) *and* that affected treatment and control bonds differently; a necessary—but far from sufficient—condition for

that would be that foreign-law issuance correlates with issuer characteristics we cannot control for. Even though treatment and control bonds are, at least statistically, the same in observable dimensions—currency, credit spreads (as a proxy for default risk), maturity, and rating in the specifications with risk controls (see Table 2)—they still could, in principle, differ in unobservable ones, notably liquidity. In that case, coincident shocks to the market price of liquidity risk could spuriously generate our results.

That seems especially far fetched for the NY-law experiment, since it comprises a sequence of fifteen events, but perhaps not impossible to imagine for the English-law experiment, since it comprises just one. Thus, for that experiment, we do two data-driven placebo exercises to provide further assurance that what we estimate is due to the ruling.

The first, a “time-series placebo,” shows, by brute force, that events that could generate spurious results are rare in our sample. We assign the event dummy to each of the 646 days in our sample period, excluding dates $t - 1$ to $t + 1$ for the baseline event and the other ruling described in Section 5. We re-estimate regression (R1) and construct an empirical distribution of the estimated t -statistics of interest for these placebo events. We then calculate “placebo p -values” for the results, each counting the sample days on which we observe differences in credit spread changes between treatment and control groups that produce larger effects (as measured by their t -statistics) than those obtained for the event date. Table 4 reports the results.

These placebo p -values should be interpreted as (an estimate of) an *upper bound* on how likely our results are to occur by chance. The bound need not be tight as the results could reflect not only “noise,” but also systematic events on other days in the sample, especially given the treatment and control bonds are not identical in some dimensions, such as risk in the unmatched specification or region in the matched specification for the direct effect. Indeed, absent risk controls, effects like ours, while uncommon, are not exceedingly rare with respect to conventional confidence levels: The placebo p -value is 7.1% for the direct effect, and 13.8% for the indirect effect. Jointly, however, the effects are rare, even without risk controls, and occur on only 2.2% of sample days.

Using the matching specification to control for risk differences tightens the bounds substan-

tially: The placebo p -value is 7.9% for the direct effect (where we cannot match within region so that the placebo is vulnerable to region-specific shocks), falls to 2.9% for the indirect effect, and to 0.2% for the two jointly—a set of direct and indirect effects like ours happens on only one other day in the sample.

The second test, a “cross-sectional placebo,” is based on the idea that, in the unlikely circumstance that a confounding event did occur in the event window, it would probably have been something that affected local-law bonds differently from *all* foreign-law bonds, not just English-law ones. Hence we test whether (i) bonds governed by New York law (direct placebo) or (ii) bonds issued by countries which have such bonds outstanding (indirect placebo), react to the English-law ruling (controlling for the true treatment effect by including all regressors from (R1)). We use the specification without risk controls because, first, it stacks the deck in favor of our finding an effect and, second, the matching specification, in which NY-law bonds are direct controls for EN-law bonds, is infeasible. As reported in Table 4, we find “nothing,” i.e. the estimates are small, both statistically and relative to the analogous baseline estimates in columns 1 and 4 of Table 3, and some of them have the “wrong” sign.

Our NY-law experiment comprises fifteen events with eight positive and seven negative shocks to commitment. We think that makes confounded estimates unlikely, as that would require multiple coincident shocks; however, you could worry that a single, large coincident shock might bias results. To test this and other, generic concerns about the robustness of the NY-law results, we split the sample into positive versus negative shocks, that is, rulings against versus in favor of Argentina. This inevitably comes with a loss of power, but columns (7) and (8) of Table 5 show that the point estimates for both effects are (i) quantitatively similar across the two subsamples, being roughly equidistant from the full-sample estimates in Table 3, and (ii) within the margins of statistical significance despite the loss of power, with respective p -values of 0.030, 0.100, 0.103, and 0.115.

There is another concern that could lead to spurious results that pertains to (and only to) our matching estimate of the indirect effect in the NY-law experiment: That sample tilts heavily towards Italian (treated) and Spanish (control) bonds, so country-specific noise could

generate a false positive. But our other specifications, which are not subject to that criticism, generate similar results for that effect. Moreover, that estimate is based on a series of fifteen events—some increasing commitment, others decreasing it so that any confounding shock would have to match not only the timing but also the direction of those fifteen events.

3.4.2 Concern: Exclusion Restriction

A general concern is that our experiments cause our results through a channel other than increasing commitment not to restructure and its associated spillover effect.

One possibility is that the experiments affected treated bonds other than via governing law. That is unlikely for the English-law experiment, in which the defendant was a firm, not a country. But it could be conceivable for the NY-law experiment, which changed not only how hard it was to restructure NY-law bonds, but also how likely Argentina was to default. Thus, as mentioned in Section 3.2, to say that our results follow from a change in commitment not to restructure, we must make an additional assumption (the exclusion restriction): Countries' propensity to borrow under NY law is not conditionally correlated with their exposure to Argentina's default. This seems like a weak assumption, in light of our matching and our region and risk controls, but we provide an additional test by removing bonds issued by South American countries from the sample. We report the results in Table 5. Our findings are robust.

Another, albeit more remote, concern is that the rulings under foreign law spill over onto local law by setting international precedents. Courts' use of foreign precedents is limited (e.g., Simon, 2013) and, to the extent that it happens, it would pose a problem only to the extent that countries that issue under one foreign (English or NY) law are systematically more likely to follow precedents from that law. Either way, we can use our cross-sectional placebo analysis to rule out such spillovers, at least in the case of English and NY law. As discussed in Section 3.4.1, the English-law experiment had no effect on NY-law bonds. An analogous placebo test shows that the NY-law experiment had no effect on English-law bonds (also in Table 4). We think such potential precedent-based spillovers are most likely between English and New York law, as they are the main foreign laws countries issue under. Their absence reassures us that

such spillovers to other laws are unlikely.

3.4.3 Concern: Market Inefficiency and Measurement Error

We use a narrow event window in our baseline because it makes it hard to imagine violations of our identifying assumptions (Section 3.2). But it could also make it easier to imagine that (i) some bond prices reflect short-term market inefficiencies, like overreaction or fire sales, or (ii) some bonds may be illiquid and not trade over the window at all, so quoted prices might not be trading prices (as most of the bonds trade OTC, we cannot observe volumes). In the former case, our results could overstate the true effect of the experiment. In the latter, they would understate it. To address concerns about such market inefficiencies or measurement error, we double the event window to four trading days (plus an intermittent weekend) for the English-law experiment. This mitigates (i) by allowing for inefficiencies to work themselves out and (ii) by making it unlikely that a bond does not trade at all over the event window. (Such a test is, unfortunately, impossible for Argentina, as many of the events in the experiment are within a few days of each other or of potentially confounding events excluded for other reasons (Hébert and Schreger, 2017).) We report the results in Table 5. Our findings are robust.

Fire sale spillovers could be the most acute concern for the NY-law experiment, given its immediate effect on Argentine bond prices. Although these bonds are dropped from our sample, they could, in principle, induce spillovers via investor wealth effects. Two facts all but rule out that this channel could generate our results. (i) The English-law experiment is immune to this concern and produces direct effects consistent with the New York-law experiment. (ii) The effect predicted by a standard model would go in the opposite direction: The rulings that decrease Argentine bondholder wealth are the rulings against Argentina; those are associated with falling yields on (other) NY-law bonds relative to the control, inconsistent their being fire sold.

You could also worry that wealth spillovers from directly treated bond are the source of the indirect effect. Absent portfolio data, we cannot test this directly. So, in the spirit of Indarte (2024), we can use the NY-law experiment to test another likely prediction of the

mechanism: the spillovers are likely larger for losses than for gains. Table 5 already provides some evidence against this, showing similar results for positive and negative shocks. But we can also do a more formal test by augmenting the baseline specification (R1) with variable $\text{Sign}_t^{AR} := \mathbb{1}_t^{Event} \text{sgn}(\Delta P_t^{AR})$, i.e. a variable that equals one at an event date signed by the change in Argentina’s default probability, and zero otherwise. Given that the average treatment effect is negative—i.e. increases in commitment lower yields—a negative coefficient on the triple interaction $\Delta \text{Comm}_t \times \mathbb{1}_t^{\text{Indirect}} \times \text{Sign}_t^{AR}$ would indicate that the indirect effect is stronger following yield-increasing shocks, consistent with a wealth channel. Instead, across all three baseline specifications—without risk controls, with rating-time fixed effects, and with matching—the estimated coefficient is positive but statistically indistinguishable from zero.

4 Model and Quantification

Here we develop a model of sovereign default/restructuring (Section 4.1). We use it to investigate the mechanism behind the empirical findings (Section 4.2), to interpret their magnitudes (Section 4.3), and to calculate elasticities for a (back-of-the-envelope) policy counterfactual (Section 4.4).

4.1 Model Set-up

We consider a one-period model of a sovereign debtor. At the end of the period, it generates random output and its outstanding bonds come due. We denote the distribution function of output under the risk-neutral ($\mathbb{Q}$) measure by $F(z)$ and its density by $f(· := F')$.

The model has two key ingredients.

- The sovereign has two different types of bonds. It has “rigid bonds” with face value D_r and “flexible bonds” with face value D_f .
- The sovereign has a willingness-to-pay problem.²⁵ It has the option to default strategically

²⁵On the importance of the problem, see Reinhart and Rogoff (2009), who say

If the reader has any doubt that willingness to pay rather than ability to pay is typically the main determinant of country default, he or she need only [observe] that more than half of defaults by middle-income countries occur at levels of external

on either one type or both types of debt. But default destroys a fraction of output;²⁶ the remaining surplus is divided with any bondholders not made whole. We denote default costs by c_i and (a scaled version of) bondholders' bargaining power by ω_i , as specified in the default payoffs below.

In our empirical environment, rigid bonds correspond to foreign-law bonds and flexible bonds to local. From Section 2.1, the haircut on rigid debt satisfies $h_r = 1 - \pi_r$, where π_r is the probability that a hold-out wins in court. Our English-law experiment, upholding consent payments, lowered π_r , raising h_r —a negative shock to commitment; our NY-law ones, striking down Argentina's *pari passu* threats, raised π_r , lowering h_r —a positive shock. In the model, ΔComm_t thus corresponds to a change in h_r , captured by changes in ω_r .

The sovereign chooses among three actions, denoted by $a \in \{\emptyset, i, rf\}$ and defined as follows:

- Full repayment ($a = \emptyset$): i debt gets D_i and the sovereign gets what is left:

$$\Pi^\emptyset = z - D_r - D_f. \quad (3)$$

- Selective default on i debt ($a = i$): D_{-i} is paid in full and output is reduced to $(1 - c_i)z$.

The sovereign gets fraction $1/(1 + \omega_i)$ of what is left:

$$\Pi^i = \frac{1}{1 + \omega_i} \left((1 - c_i)z - D_{-i} \right). \quad (4)$$

- Cross-default ($a = rf$): Output is reduced to $(1 - c_r)(1 - c_f)z$ and the sovereign gets a fraction $1/(1 + \omega_r + \omega_f)$ of what is left:

$$\Pi^{rf} = \frac{1}{1 + \omega_r + \omega_f} (1 - c_r)(1 - c_f)z. \quad (5)$$

If the sovereign defaults on i debt, then i debt takes a haircut, denoted by h_i^a . The haircut

debt relative to GDP below 60 percent, when, under normal circumstances, real interest payments of only a few percent of income would be required to maintain a constant level of debt relative to GDP (p. 54).

²⁶Such costs can capture exclusion from financial markets (Eaton and Gersovitz, 1981) or trade sanctions (Bulow and Rogoff, 1989).

is determined such that i -bondholders get a fraction ω_i of what the sovereign gets:

$$h_i^a = \begin{cases} 0 & \text{if } a = \emptyset, \\ 1 - \frac{1}{D_i} \frac{\omega_i}{1 + \omega_i} \left((1 - c_i)z - D_{-i} \right) & \text{if } a = i, \\ 1 - \frac{1}{D_i} \frac{\omega_i}{1 + \omega_r + \omega_f} (1 - c_f)(1 - c_r)z & \text{if } a = rf. \end{cases} \quad (6)$$

Normalizing the face value of each of the sovereign's bonds to one (so D_i is the quantity of i bonds outstanding) and assuming competitive pricing, we can express the price of a bond of type $i \in \{r, f\}$ as

$$p_i = 1 - \mathbb{E}^{\mathbb{Q}}[h_i^a], \quad (7)$$

where we use $\mathbb{E}^{\mathbb{Q}}$ to denote the expectation under the risk-neutral measure and have the price inflated by the risk-free rate for ease of notation.

For simplicity, we assume that, r debt being rigid, defaulting on it selectively is dominated; so rigid debt is never selectively renegotiated.²⁷

The sovereign now chooses its action a to maximize Π^a over $a \in \{\emptyset, f, rf\}$; see equations (3)–(5). As illustrated in Figure 2, there are two regimes, which we label “selective default” and “cross-default.”

- Selective default regime: Here there are two cutoffs $z^{f \succ rf}$ and $z^{\emptyset \succ f}$ such that if $z < z^{f \succ rf}$ the sovereign defaults on both r and f debt; if $z^{f \succ rf} \leq z < z^{\emptyset \succ f}$, it defaults selectively on f debt ($a = f$); otherwise, if $z \geq z^{\emptyset \succ f}$, it repays in full ($a = \emptyset$).
- Cross-default regime: Here there is one cutoff, $z^{\emptyset \succ rf}$, such that if $z < z^{\emptyset \succ rf}$, it defaults on both types of debt and, otherwise, if $z \geq z^{\emptyset \succ rf}$, it repays in full.

²⁷That local-law bonds are typically easier to restructure than foreign- is consistent with bond prices (Chamon, Schumacher, and Trebesch, 2018) and practice (e.g., the Argentina saga and Greek restructuring; see also Buchheit and Gulati (2018)). That can arise endogenously here: Comparing the payoffs in equation (4) gives sufficient conditions for $a = r$ never to be optimal: $(1 + \omega_r)(1 - c_f) > (1 + \omega_f)(1 - c_r)$ and $(1 + \omega_r)D_r < (1 + \omega_f)D_f$.

Per Figure 2, which regime obtains depends on whether $z^{\emptyset \succ rf} \leq z^{f \succ rf}$: Either way, the set-up probably understates the likelihood of cross-default, which could arise not only for the strategic reasons underscored here but also for other reasons, like pari passu and cross-default clauses in bond indentures.²⁸ That squares with our empirical results, as we discuss below.

4.2 Model Results

Here we consider the comparative statics corresponding to our experiments, interpreted as an increase in the bargaining power of rigid debt ω_r .

First we consider the effect on rigid debt itself:

Proposition 1. *The price of rigid (r) bonds is increasing in rigid bondholders’ bargaining power ω_r : $\partial p_r / \partial \omega_r > 0$.*

Increasing ω_r has two effects on r debt. First, almost by definition, it increases what r debt gets in the event of default. That also symmetrically decreases what the sovereign gets, leading to the second effect: It makes default less attractive to the sovereign, making it more likely that it repays in full. Both effects benefit rigid bondholders, so the price of rigid bonds unambiguously rises. That is consistent with our findings on the direct effect of our experiments (Section 3.3.1).

We now turn to the spillover effects on f debt:

Proposition 2. *The price of flexible (f) bonds can be increasing or decreasing in rigid (r) bondholders’ bargaining power ω_r . Necessary and sufficient conditions for their signs are given in the proof.*

The mechanism depends on the regime. In the selective default regime, the bargaining power of r debt has no effect on the probability of repaying f debt—the cutoff $z^{\emptyset \succ f}$ does not depend on ω_r . The only effect is that it decreases the payment when the sovereign defaults on all debt,

²⁸Pari passu clauses may preclude payments to one class without payments to another and cross-default clauses say a default on one class constitutes a default on another (Choi, Gulati, and Posner, 2012). Cross-default has recently been a requirement for the IMF, which set a new precedent by saying it would support restructuring one class of Ghanaian debt only if the country restructured another as well; see, e.g., “Ghana Default Puts Domestic Debt ‘Can of Worms’ in the Spotlight” (*Financial Times*, April 13, 2023).

as r debt’s higher bargaining power helps it extract surplus from f debt. That is a burglar alarm effect akin to that in Bolton and Jeanne (2007).

In the cross-default regime, the default threshold $z^{\emptyset \succ rf}$ does depend on ω_r , leading to a new effect: Increasing ω_r makes default less attractive to the sovereign, making it more likely to repay all f debt. That has a positive spillover on f debt. The spillover can dominate. That is what we term the crime watch. It explains the positive spillovers in our empirical analysis.²⁹

Our framework here nests the sovereign restructuring models in Bolton and Jeanne (2007) and Carletti et al. (2020), *mutatis mutandis*. Those papers have two types of debt, flexible debt (termed “renegotiable”) that can be restructured costlessly and rigid debt (termed “non-renegotiable”) that effectively cannot be restructured at all ($c_r = 1$)—there is only outright default on rigid debt.³⁰ Those assumptions ensure that the burglar is the only spillover. They do not admit our crime watch and, as such, do not explain our empirical findings. Moreover, they preclude the typical sovereign default, which is a costly restructuring with positive creditor recovery (Cruces and Trebesch, 2013).

We go further in two directions, by allowing for (i) renegotiation of both types of debt, with potentially different costs, and (ii) general bargaining power $\omega_i \geq 0$. Thus our model admits both the burglar alarm and crime watch. The burglar alarm arises for the same reason as in Bolton and Jeanne (2007). In the selective default regime, making one type of debt harder to restructure deflects default onto the other type. The crime watch, whereby making one type of debt hard to restructure deters default altogether, arises in the cross-default regime, which

²⁹Our finding that the cross-default regime is empirically relevant also informs the model parameters. Cross-default requires $z^{\emptyset \succ rf} < z^{f \succ rf}$ (see Appendix C.1.1), which simplifies to

$$\frac{D_f}{D_r} < \frac{\omega_f(1 - c_r) + c_f(1 + \omega_f + \omega_f c_r)}{(1 - c_f)(\omega_r + (1 + \omega_f)c_r)}.$$

That is satisfied whenever c_f is sufficiently large, underscoring that restructuring flexible debt is not costless—one of our main departures from Bolton and Jeanne (2007) and Carletti et al. (2020). This is also mutually consistent with the no-selective-default-on- r conditions in footnote 27 as long as, additionally, $\omega_r > \omega_f$ and $c_r > c_f$, both of which we think are natural in our setting: Foreign-law (Rigid) creditors have stronger legal protections (high ω_r) and defaulting on them is more likely to trigger cross-default clauses and litigation (high c_r).

³⁰For them, the bargaining power of rigid debt is immaterial, because default destroys everything; the bargaining power of flexible debt is our $\omega_f/(1 + \omega_f)$.

is absent in Bolton and Jeanne (2007). Our empirical findings suggest it could be relevant in practice.

Our model can also help explain why the empirical literature is divided on what type of debt trades at a premium. Whereas Chamon et al. (2018) find foreign-law (rigid) bonds trade at a premium relative to local-law (flexible), Carletti et al. (2020) find that bonds with CACs (flexible) trade at a premium relative to those without (rigid). Our model reconciles those *prima facie* conflicting findings. Per equation (7), if selective default is sufficiently rare and $\omega_r/D_r < \omega_f/D_f$, then flexible debt carries the premium ($p_f > p_r$); otherwise, rigid debt does ($p_r < p_f$). Moreover, the empirical settings in Chamon et al. (2018) and Carletti et al. (2020) seem broadly consistent with the parameter regimes that generate the corresponding results in our model.³¹

4.3 Implied Haircut Changes

The model can also help us to interpret our experiments, which do not immediately lend themselves to quantification—the NY-law rulings made restructuring harder on net and the English-law rulings made it easier, but by how much did they affect the haircuts h_r ? Here we show how to use the model to back this number out from our empirical estimates and a few other largely observable sufficient statistics.

Given it explains our reduced-form results, we focus on the cross-default regime. There is one relevant cutoff, which we abbreviate by $z^* := z^{\varnothing > rf}$. Using a first-order approximation, we derive a system of equations for the changes in spreads on r - and f -bonds, denoted by Δs_r and Δs_f , in terms of the change in haircut on rigid debt induced by the experiments, denoted by Δh_r :

³¹The condition for $p_r > p_f$, that $\omega_r/D_r > \omega_f/D_f$, corresponds to the local-law advantage in Chamon et al. (2018). Symmetrically, the condition for $p_f > p_r$ is a possibility for the CAC bonds in Carletti et al. (2020), as CACs, by construction, facilitate bondholder coordination (ω_f is high). Moreover, the CAC in the sample features rigid dual-limb voting aggregation, which preserves some hold-out power for individual bondholders (Tinti, 2020; Galli and Guibaud, 2025). In contrast, the local-law bonds without CACs are susceptible to cram-downs as in the Greek restructuring (Zettelmeyer et al., 2014).

Proposition 3. *A small change in the haircut on rigid bonds Δh_r , induced by a small shift in bargaining power ω_r , translates to changes in the spreads of r - and f -bonds, Δs_r and Δs_f , approximately as follows:*

$$\Delta s_r \approx \frac{1}{p_r T_r} \left(F_{h_r} h_r + F(z^*) \right) \Delta h_r, \quad (8)$$

$$\Delta s_f \approx \frac{1}{p_f T_f} \left(F_{h_r} h_f - F(z^*) \frac{1 - h_f}{1 - h_r} \frac{\omega_r}{1 + \omega_f} \right) \Delta h_r, \quad (9)$$

where T_i is the maturity of bonds of type $i \in \{r, f\}$ and F_{h_r} is short-hand for $\frac{\partial}{\partial h_r} F(z^*(h_r))$.

Each equation has two terms, capturing deterrence and recovery. In equation (8), both work in the same direction for rigid bondholders: More commitment lowers the probability of default and improves recovery conditional on default. In equation (9), the two terms are spillovers that go in opposite directions: The deterrence term is the crime watch—lower default probability benefits flexible bondholders too—while the recovery term is the burglar alarm—in cross-default, rigid creditors capture a larger share of the surplus, leaving less for flexible ones. Which one dominates determines the sign of the spillover.

We solve the system for countries with both directly and indirectly treated bonds outstanding. As we detail in Appendix D, we use our estimates of Δs_r and Δs_f from our matching specifications, the five-year CDS premiums from Markit to calculate $F(z^*)$, five-year bond maturities T_r and T_f to match the CDS, the model-implied haircuts from equation (7) for h_r and h_f , and yields adjusted for risk-free rates and currency risk from our data. These quantities are not quite sufficient statistics to solve for Δh_r . We also need the relative bargaining weight $\omega_r/(1 + \omega_f)$. But, as we will see below, our quantification turns out to be robust to allowing for a wide range of reasonable values for it. We do not need to know the costs of default c_i or the sensitivity of the default probability to the r -haircut, F_{h_r} —in fact we back that out too; see the next section.

We solve the system (8)–(9) with $\frac{\omega_r}{1 + \omega_f} \in [0.01, 1]$. The smaller values are probably more plausible, as creditor recoveries are small relative to output z (cf. footnote 25), correspond-

ing to $\omega_i \ll 1$. For a concrete, albeit anecdotal calibration, we use the March 2012 Greek restructuring with data from Zettelmeyer et al. (2014). Restructured face values were 177.3 billion EUR in local-law (flexible) bonds and 20 billion in English-law (rigid) bonds, with an estimated fair-value haircut of 64.6%. The haircut applied to effectively all local-law bonds in the restructuring, but to only 71% of English-law bonds, with the remainder repaid in full (implying a 46% effective haircut on rigid debt). We use Greek GDP, which was 193 billion EUR in 2012, as a proxy for output after restructuring costs, $z(1 - c_f)(1 - c_r)$. These values imply a recovery of 10.8 billion EUR for rigid creditors, 62.8 for flexible creditors, and 119.4 for the borrower. Thus, normalizing total debt to one, $\frac{\omega_r}{D_r} = 0.89$ and $\frac{\omega_f}{D_f} = 0.58$, consistent with a relatively weak bargaining position for local-law creditors (i.e. the local-law advantage). Applying these estimates to the English- and NY-law samples (where the rigid-to-flexible debt ratios differ slightly) implies $\frac{\omega_r}{1+\omega_f} = 0.13$ for English law and 0.09 for New York law.

We estimate that haircuts increased by 33 bps (calibrated to Greece, 18-37 bps using the wider range) from the High Court ruling and moved by around 25 bps (12–27 bps) for each event in the Argentina saga on average. These magnitudes accord with our perception of the events as mattering for specific restructuring instruments but not revolutionizing market practice. We think that affords some validation that our model, despite being highly stylized, could be quantitatively relevant, or at least that the framework could be a useful building block for future (less highly-stylized) quantitative work.

4.4 The Default-to-haircut Elasticity and a Policy Counterfactual

We can also use the model to estimate by how much an increase in haircuts increases the default probability, a potentially relevant input for sovereign debt modeling and policy analysis. As many models and policies do not distinguish between r - and f -bonds, we work with the average haircut here: $\bar{h} := \varphi h_r + (1 - \varphi) h_f$, where $\varphi := D_r / (D_r + D_f)$. The next result describes how to use observable quantities to translate F_{h_r} , which we can back out from the system in Proposition 3, into the desired elasticity:

Proposition 4. *The elasticity of the default probability $F(z^*)$ with respect to the average haircut $\bar{h}$ is*

$$\mathcal{E} = \frac{\bar{h}F_{h_r}}{F(z^*)} f \frac{(1 - h_r)(1 + \omega_f)(\omega_r + \omega_f)}{\omega_r(1 - \bar{h})} \quad (10)$$

where F_{h_r} solves the system in Proposition 3.

Applying formula (10) to averages from the raw data, bargaining-power estimates from Greece, and the solution of the system (8) and (9), we calculate $\mathcal{E}$ of 17.2 for English-law issuers and 13.4 for NY-law issuers. I.e. a 1% increase in haircuts increases the risk-neutral default probabilities by around 17.2% for English law and 13.4% for NY (corresponding to 3.0 and 1.4 percentage points, respectively). These numbers are broadly in line with existing estimates from the literature on household bankruptcy, where haircuts (viz. asset exemptions) have received relatively more attention than in the sovereign default literature.³² That commitment benefits English- more than NY-law issuers likely reflects that English-law issuers have higher debt and lower default costs, which, through F_{h_r} , feeds into the expression for $\mathcal{E}$.

To illustrate how $\mathcal{E}$ can be used for policy counterfactuals, we use the NY-law number to approximate how a change in haircuts changes the expected deadweight costs of default for Argentina, for which we have an estimate from Hébert and Schreger (2017). To do so, we define the total default cost under cross-default, $c := c_r + c_f - c_r c_f$, and use the Euler approximation to write

$$\Delta \text{Q-expected cost} = \Delta \int_0^{z^*} cz dF(z) \approx \frac{cz^* \mathcal{E}}{\bar{h}/F(z^*)} \Delta \bar{h}, \quad (11)$$

having used the fundamental theorem of calculus and substituted for $\mathcal{E}$ from above. Replacing cz^* with Hébert and Schreger’s estimate, $\mathcal{E}$ with ours from the NY-law experiment, and the other parameters with data on Argentina, we find that a one percentage point increase in haircuts saves around 4.9 billion USD in deadweight costs of default. (See Appendix D.4 for omitted details from this section.)

³²Fay, Hurst, and White (2002) find that a \$1000 increase in the payoff in bankruptcy increases filings by around 7%. That \$1000 is around 0.7% of the net debt discharged in that sample, making the elasticity 10—slightly below the estimates from our two experiments.

5 External Validity: An Additional Event

We interpret our results broadly—bondholders’ bargaining power in restructuring disciplines a sovereign debtor. They are based on two distinct experiments, covering bribes and threats under English and NY laws. But external validity can always be a question. To address it further, we consider an additional experiment that made it harder to restructure English-law debt (Section 5.1). Although it presents greater identification challenges than our baselines (Section 5.2), it affirms its message: The commitment not to restructure one class of bonds benefits that class as well as others.

5.1 Assenagon Asset Management v Irish Bank Resolution Corp

Here we exploit a ruling pertaining to another controversial restructuring device: exit consents, which are the stick to consent payments’ carrot. When tendering creditors part with their bonds (“exit”), they agree (“consent”) to changes in the terms of other bonds in the same class. Like the other threats and bribes, they counter the hold-out problem, but can create a hold-in problem, and therefore be deemed coercive.³³

Just two months after its ruling on consent payments in *Azevedo v Imcopa* (Section 2.2), the High Court ruled on exit consents.³⁴ The case was brought against Anglo Irish by a bondholder, Assenagon Asset Management, who held out from an exchange offer. After having received just €170 for Anglo-Irish bonds with face value of €17 million(!), Assenagon sued, submitting, *inter alia*, that the exit consents in the exchange offer were abusive. The presiding judge, Michael Briggs, agreed, ruling on July 27, 2012 that the restructuring was illegal. It was the Court’s first decision on exit consents and thus, as noted in the decision, “of wide importance within the bond market.” It made restructuring all English-law bonds harder, thus increasing debtor commitment.

³³They are nonetheless alive and well in the sovereign debt market broadly, as evidenced by a recent proposal to restructure Lebanese debt, namely Luo, Smith, and Xiao (2020).

³⁴See *Assenagon Asset Management SA v Irish Bank Resolution Corp Ltd* (formerly *Anglo Irish Bank Corp Ltd*) [2012] EWHC 2090 (Ch) (27 July 2012).

5.2 Estimation and Identification Challenges

We run the same regressions as in our baseline specifications (R1), setting the treatment variable ΔComm_t equal to one on the trading day after the ruling, that is, on July 30, 2012.

We mitigate identification concerns in the same way as in the baseline (Section 3.2). Thus, as above, to be a threat, an event must happen at nearly the same time as the experiment and affect both treatment and control bonds differently within the same currency, maturity, and region/risk category.

Unlike above, such a threat is more than a remote possibility as the ruling could be confounded by a speech that ECB President Mario Draghi gave on the day before the ruling, in which he promised to do “whatever it takes to preserve the euro,” a statement that could have affected treated and control bonds differently.

Although this threat makes us more tentative in our interpretation, it does not necessarily undermine our identification. Added noise notwithstanding, the experiment remains valid if we control for pricing-relevant variables that respond to the speech. As in the baseline, we control for credit risk using rating fixed effects and matched bond pairs. We further exclude bonds issued by Eurozone member countries.

5.3 Do Sovereign Creditors Benefit from Committing Not to Restructure?

Table 6 reports the estimated coefficients for three specifications, in which we do what we can to avoid bias due to Draghi’s confounding speech. The first (column 1) is the baseline matching specification, which compares only similar bonds (i.e. presumably similarly affected by the speech). In columns 2 and 3, we modify that specification and the one with ratings fixed effects by dropping all Eurozone bonds (for the direct effect only in the case of matching, because there are no non-Eurozone bonds in the matching specification for the indirect effect). The signs of the estimates on all coefficients of interest in all specifications line up with the baseline experiments: Increasing commitment leads spreads on both directly and indirectly treated bonds to decrease.

The fixed-effects estimate for the indirect effect is statistically significant with a p-value of 5%. For the direct effect, the matching estimates with and without Eurozone bonds have p-values of 6% and 10%, respectively. Overall, this exercise lacks statistical power relative to the baseline, likely due to (i) noise and/or confounding effects introduced by Draghi’s speech and (ii) the low number of observations, especially for those specifications without Eurozone bonds. The non-Eurozone sample of 741 bonds in the fixed-effects specification includes few directly treated bonds (161) relative to indirectly treated ones (400, plus 180 control bonds), and, accordingly, the direct effect is statistically weaker in this specification than the indirect one. However, the direct effect does have some statistical support in both matching specifications.

Overall, Draghi’s speech seems to attenuate rather than inflate our estimates, given that (i) adding Eurozone bonds shrinks the estimate of the direct effect (comparing column 2 with column 1), and (ii) the matching estimate of the indirect effect in column 1, which is based entirely on Eurozone bonds and thus most vulnerable to confounding effects, is close to zero. That suggests that noisy Eurozone bonds seem more likely to generate false negatives than false positives, possibly allaying concerns that any of our baseline results could be false positives due to events unfolding during the European sovereign debt crisis.

6 Conclusion

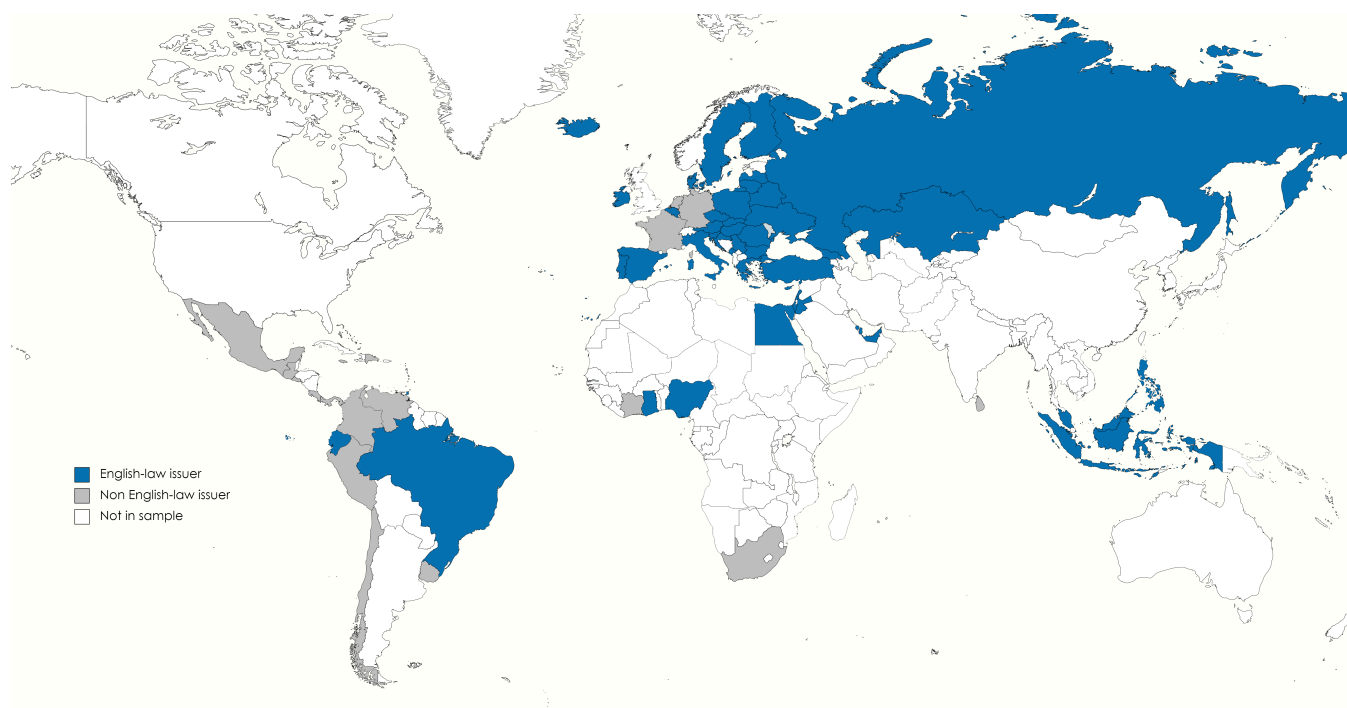
We bring evidence from natural experiments to bear on the value of commitment not to restructure sovereign bonds. Our research design allows us to distinguish between the value for bondholders individually and collectively, and thereby assess cross-bond spillovers, like those implied by the “burglar alarm” view that to commit not to restructure one bond is only to encourage the restructuring of others (Bolton and Jeanne, 2007, 2009).

Our empirical results suggest a perspective new to the literature. We find that committing not to restructure one bond is more like a successful neighborhood crime watch than a self-defeating burglar alarm, viz. it benefits both that bond and others as well. We develop a model of the “crime watch” perspective that not only formalizes the mechanism behind it

but also proves useful to rationalize other empirical results in the literature, quantify natural experiments, and analyze policy counterfactuals.

Figure 1: **Treated and control countries.** Countries in the treatment group for the shocks to English law and New York law are in blue/red. Countries in gray are in the control group. Countries in white are not part of the sample.

Panel A. English-law experiment



Panel B. New York-law experiment

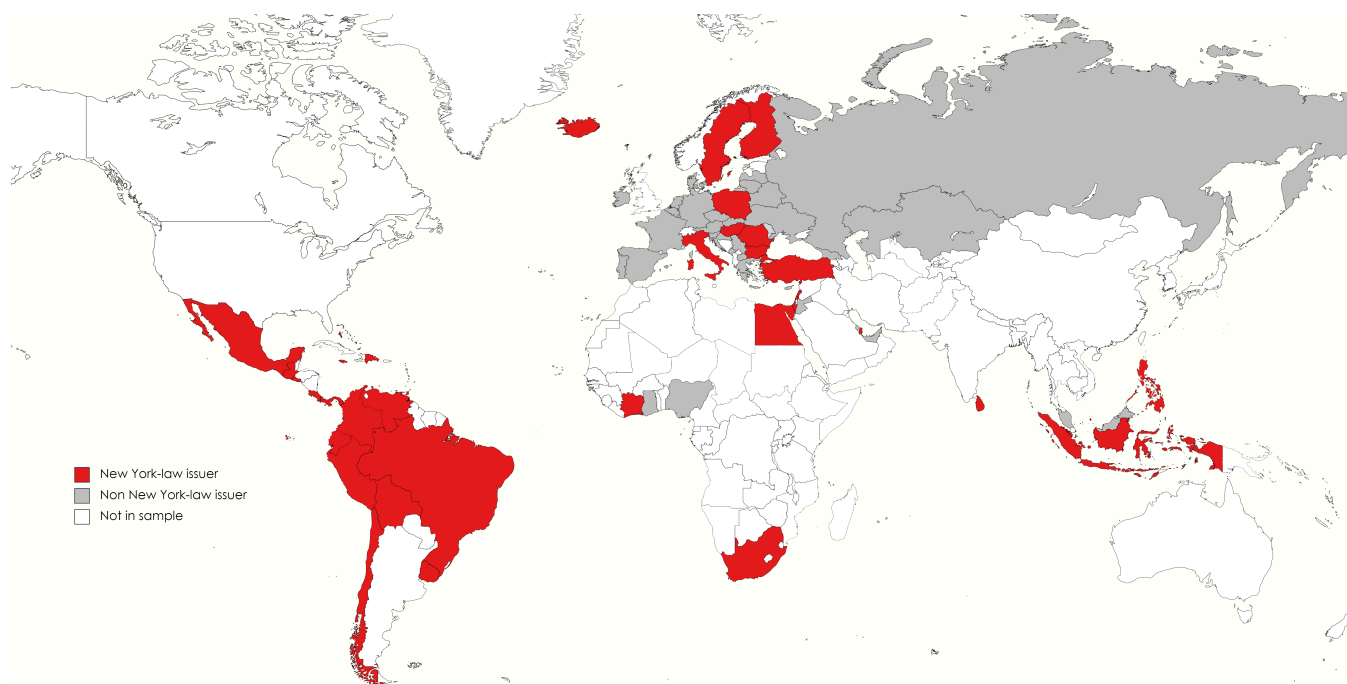
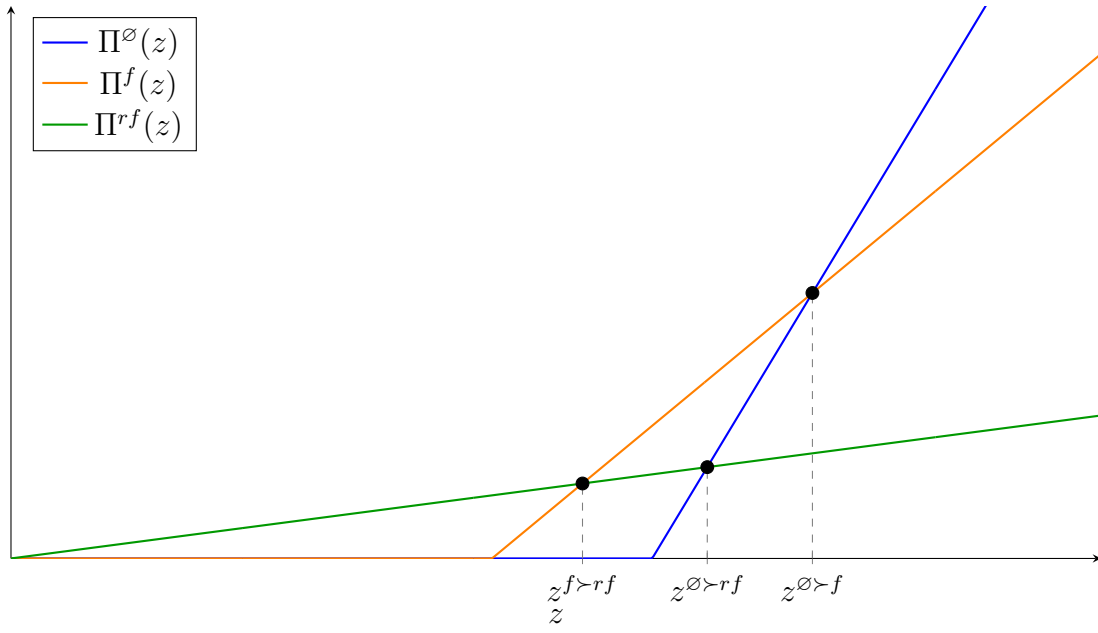
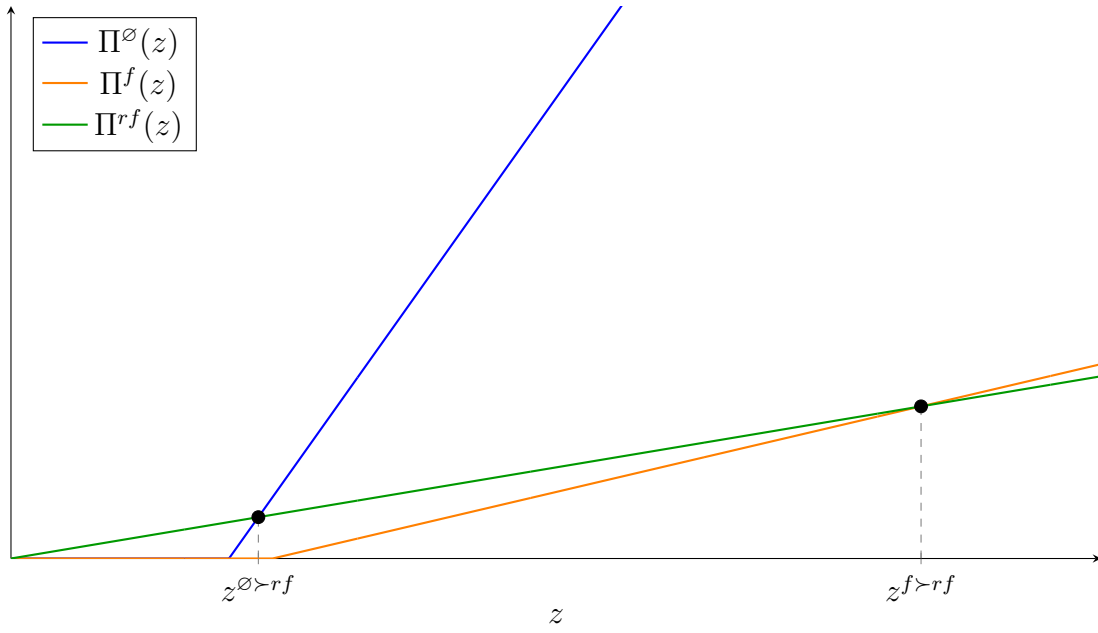


Figure 2: Sovereign payoffs



(a) Selective default regime



(b) Cross-default regime

Table 1: **Summary statistics on the event date.** We report statistics for the 67 countries with bonds in our sample on the eve of the (first) event. For the Imcopa experiment, this is Tuesday, May 29, 2012, ahead of the announcement of the ruling on May 30. For the Argentina experiment, this is Friday November, 23, 2012, ahead of the first ruling identified by Hébert and Schreger (2017). In the first three columns, we report statistics by bond group (direct treatment, indirect treatment, and control), with the total number of bonds in the first row and group-means in the remaining rows. These columns provide bond-level comparisons between the two treatment groups and the control group. The other columns report the distributions across countries of the same variables averaged by country (e.g., the average number of bonds per country is 20.48). These columns describe the country-level heterogeneity in the respective variables.

Variable	Bond-level statistics			Distribution of country means				
	Direct	Indirect	Control	Mean	St. Dev.	1%	Median	99%
English-law experiment								
Number of bonds	235	715	331	19.12	21.74	1	12	123
Face value per bond (\$bn)	1.45	2.20	3.29	1.52	1.42	0.18	0.96	6.87
Yield (%)	5.20	4.29	3.01	5.19	3.08	0.58	4.52	13.85
Credit spread (%)	3.91	2.11	1.36	3.39	3.08	-1.38	2.87	11.51
Maturity	8.17	8.22	9.81	9.15	4.63	2.20	8.05	22.22
Fraction English law	1.00	0.00	0.00	0.35	0.39	0.00	0.14	1.00
Fraction New York law	0.00	0.23	0.40	0.24	0.35	0.00	0.00	1.00
Fraction local law	0.00	0.73	0.59	0.39	0.37	0.00	0.29	1.00
New York-law experiment								
Number of bonds	311	521	668	19.49	21.19	1	14	116
Face value per bond (\$bn)	1.13	3.89	2.62	1.55	1.58	0.18	1.00	8.37
Yield (%)	3.61	3.62	3.06	4.23	2.84	0.53	3.84	13.85
Credit spread (%)	2.47	1.01	1.68	2.80	2.82	-1.55	2.11	11.51
Maturity	10.74	7.93	7.56	9.16	4.49	1.71	8.27	21.74
Fraction English law	0.00	0.15	0.26	0.34	0.39	0.00	0.14	1.00
Fraction New York law	1.00	0.00	0.00	0.25	0.36	0.00	0.00	1.00
Fraction local law	0.00	0.81	0.73	0.39	0.37	0.00	0.28	1.00

Table 2: **Balance tests on matched bond pairs.** We report regressions of different bond or issuer characteristics on the treatment indicator for bonds involved in the matching estimation. Characteristics are measured on the day before the first event date for each experiment. Credit spreads, as a proxy for default risk, are expressed in percentage points, maturity in years, and rating in notches, with one notch being the difference between, for instance, AA and AA−. Standard errors are clustered by country and reported in parentheses. Asterisks denote significance at the 1, 5, and 10% significance level, respectively.

	Default risk	Maturity	Rating	Debt/GDP
Treated	0.157	-0.054	0.169	0.110**
	(0.201)	(0.108)	(0.157)	(0.054)
Bonds	924	924	924	838

Table 3: Baseline estimates: The effect of the ruling on directly treated and indirectly treated bonds. We estimate the DiD regression (R1) (Section 3.3). We report standard errors in parentheses, clustered at the country level, and at the country-match level for matching specifications to account for sampling with replacement of control bonds. Asterisks denote significance at the 1, 5, and 10% significance level, respectively. For the English-law specifications, $\Delta\text{Comm}_t = -1$ on May 31st, 2012, and zero otherwise. For the NY-law specifications, $\Delta\text{Comm}_t = \Delta P_t^{\text{AR}} \mathbb{1}_t^{\text{Event}}$, that is, the change in the CDS-implied, risk-neutral default probability for Argentina, multiplied by an event indicator equal to one for each of the 15 dates corresponding to the end of the two-trading day event window around rulings in the Argentine restructuring saga catalogued by Hébert and Schreger (2017). The dependent variable—the bond-level credit spread—is expressed in percentage points. Since the dependent variable is a change over the event window(s) and all fixed effects are interacted with time, the estimation of the coefficients of interest is entirely cross-sectional. It is therefore not the number of bond-time observations, but the number of bonds (or bond-pairs for matching specifications) that matters for power, and that is what we report as the number of observations. The regressions include a longer time series of observations which are used to estimate the coefficients on the group-indicators (β_3 and β_4 in (R1)), that is, to de-trend credit spread changes by group over the sample.

	English law			New York law		
Risk control	None	Rating FE	Matching	None	Rating FE	Matching
Direct $\times \Delta\text{Comm}_t$	-0.112*** (0.037)	-0.052* (0.028)	-0.058*** (0.018)	-0.329* (0.166)	-0.245** (0.097)	-0.339** (0.148)
Indirect $\times \Delta\text{Comm}_t$	-0.133** (0.051)	-0.062** (0.028)	-0.042*** (0.009)	-0.160** (0.074)	-0.125** (0.051)	-0.160** (0.076)
Region-time	Yes	Yes	No	Yes	Yes	No
Currency-time	Yes	Yes	matched	Yes	Yes	matched
Maturity-time	Yes	Yes	matched	Yes	Yes	matched
Rating-time	No	Yes	matched	No	Yes	matched
Bond pair-time	No	No	Yes	No	No	Yes
Bonds/Matched pairs	1281	1281	207	1325	1325	251

Table 4: **Placebo tests.** We run two placebo tests. *Time-series:* We re-estimate the DiD regression (R1) for the ruling in *Azevedo v Imcopa* assigning the event indicator to 646 placebo dates. We then report in brackets the fraction of these dates for which the t -statistic of the estimated placebo-coefficients is larger than that obtained for the event window. The row labelled “Joint” reports this statistic for the joint results (β_1, β_2) . *Cross-sectional:* We estimate the placebo-treatment effect by switching the treatment law: for the English-law ruling in *Azevedo v Imcopa*, we estimate the effect on NY-law bonds (direct) and bonds of NY-law issuers (indirect) on the event date against the remaining control group. For the New York-law rulings in *Argentina v NML Capital* we estimate the effect on English-law bonds (direct) and bonds of English-law issuers (indirect) relative to the remaining control group.

	Time-series: Placebo dates		Cross-sectional: Placebo treatment	
	English law		English law	New York law
Risk control	None	Matching	None	None
Direct $\times \Delta \text{Comm}_t$	[0.071]	[0.079]	0.006 (0.047)	0.058 (0.066)
Indirect $\times \Delta \text{Comm}_t$	[0.138]	[0.029]	-0.070 (0.066)	-0.050 (0.040)
Joint	[0.022]	[0.002]		
Region-time	Yes	No	Yes	Yes
Currency-time	Yes	matched	Yes	Yes
Maturity-time	Yes	matched	Yes	Yes
Bond pair-time	No	Yes	No	No

Table 5: **Robustness tests.** We estimate the DiD regression (R1) for the English-law experiment (i) for a longer event window of four trading days (credit spread changes from $t - 1$ to $t + 3$), and (ii) in levels of credit spreads before and after the ruling with a bond fixed effect rather than two-day changes. The last four columns report a robustness test on the NY-law experiment (i) excluding all South American issuers from the sample and (ii) splitting the events by positive and negative shocks to commitment, i.e. by $\Delta\text{Comm}_t \geq 0$. We report estimates for the baseline specifications without risk controls and with matched bond pairs. Standard errors are clustered as in the respective baseline specification. Asterisks denote significance at the 1, 5, and 10% significance level, respectively.

Robustness	English law				New York law			
	4-day window		Levels		Excl. South America		+ve	-ve
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Direct $\times \Delta\text{Comm}_t$	-0.126*** (0.030)	-0.136*** (0.031)	-0.057* (0.031)	-0.055* (0.028)	-0.351** (0.172)	-0.387** (0.179)	-0.349 (0.219)	-0.315** (0.143)
Indirect $\times \Delta\text{Comm}_t$	-0.084*** (0.031)	-0.035** (0.015)	-0.125** (0.052)	-0.041*** (0.009)	-0.150** (0.071)	-0.152** (0.076)	-0.187 (0.113)	-0.140* (0.084)
Region-time	Yes	No	Yes	No	Yes	No	Yes	Yes
Currency-time	Yes	matched	Yes	matched	Yes	matched	Yes	Yes
Maturity-time	Yes	matched	Yes	matched	Yes	matched	Yes	Yes
Rating-time	No	matched	No	matched	No	matched	No	No
Bond pair-time	No	Yes	No	Yes	No	Yes	No	No
Bond	No	No	Yes	No	No	No	No	No

Table 6: **Estimates from a landmark ruling on exit consents.** We estimate the DiD regressions (R1), using a two-day event window around the ruling (July 27, 2012). We report clustered standard errors in parentheses. These are clustered at the country-level for the fixed effects specification, and double clustered by country and control bond for the matching specifications to account for repeated use of some control bonds in different pairs.

	Matching		Rating FE
Direct $\times \Delta \text{Comm}_t$	-0.051*	-0.055	-0.034
	(0.027)	(0.032)	(0.024)
Indirect $\times \Delta \text{Comm}_t$	-0.002		-0.043*
	(0.007)		(0.022)
Eurozone bonds	included	excluded	excluded
Currency-time	matched	matched	Yes
Maturity-time	matched	matched	Yes
Rating-time	matched	matched	Yes
Bond pair-time	Yes	Yes	No
Bonds/Matched pairs	207	117	741

Table 7: **Baseline estimates: The relative effect of the ruling on directly versus indirectly treated bonds within country.** We estimate the DiD regression (R1) (Section 3.3) with a country-time instead of region-time fixed effect. We report standard errors in parentheses, clustered at the country level.

	English law		New York law	
	None	Rating FE	None	Rating FE
Risk control				
Direct $\times \Delta \text{Comm}_t$	0.0032 (0.0211)	0.0031 (0.0213)	-0.0139 (0.0579)	0.0010 (0.0583)
Direct	-0.0005 (0.0004)	-0.0005 (0.0004)	0.0011 (0.0007)	0.0009 (0.0007)
Country-time	Yes	Yes	Yes	Yes
Currency-time	Yes	Yes	Yes	Yes
Maturity-time	Yes	Yes	Yes	Yes
Rating-time	No	Yes	No	Yes
Bonds	1281	1281	1325	1325

A Data

A.1 Geography

Our sample includes the following issuers, assigned to regions as listed below:

- **Europe:** Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Isle of Man, Italy, Latvia, Lithuania, Luxembourg, Montenegro, the Netherlands, North Macedonia, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden.
- **South America:** Bolivia, Brazil, Chile, Colombia, Ecuador, Paraguay, Peru, Uruguay, Venezuela.
- **Central America:** Barbados, Bahamas, Belize, Costa Rica, Dominican Republic, El Salvador, Grenada, Guatemala, Honduras, Jamaica, Mexico, Panama, Trinidad and Tobago.
- **Middle East:** Bahrain, Egypt, Israel, Jordan, Lebanon, Qatar, UAE.
- **Africa:** Côte d’Ivoire, Ghana, Nigeria, South Africa.
- **Central Asia:** Azerbaijan, Belarus, Georgia, Kazakhstan, Russia, Turkey, Ukraine.
- **East Asia:** Indonesia, Malaysia, Philippines, Sri Lanka.

A.2 Risk-Free Interest Rate Proxies

The spread at time t of bond i with yield $y_{i,t}$ maturity T , currency denomination f is defined as $s_{i,t} := y_{i,t} - r_{t,T,f}$, where $r_{t,T,f}$ is the currency-specific risk-free rate from t to T . As $r_{t,T,f}$ is unobservable, we proxy for it with the rate on the appropriate (i.e. currency and maturity matched) swap rate.³⁵

³⁵Both rates are (basically) risk-neutral expectations of the average spot rate over the maturity. The difference is the time weighting: The swap rate discounts longer maturities. If the discount rate is about constant (e.g., because maturities are short or spot rates low) then the two rates are about equal. Even if it is not, swap rate is still an effective proxy for the risk-free rate for our purposes, as it captures variation in currency-specific risk-free term-structures.

We match bonds to swap rate maturities of 1, 2, 3, 4, 5, 10, 15, 20, and 30 years. Where swap rates for the longer maturities are not available, we use the longest available maturity. Where no swap rate is available, we drop all bonds denominated in that currency from the sample. In Table SA.1 in the Supplementary Appendix, we list the different currencies in our sample along with the floating rate underlying the swap used to construct credit spreads.

B Matching Procedure

The matching procedure is as follows. For each (directly or indirectly) treated bond, we consider all untreated bonds that have (i) the same currency denomination as, (ii) maturity within one year of, and (iii) a credit rating within two notations of the treated bond's. If the set is empty, we discard the treated bond. If it is not, we choose the bond with the closest pre-event credit spread to the treated bond as its control bond. (Note that we do not exclude matching multiple treated bond with the same control bonds, i.e. we “match with replacement.”) In Table SA.4 and Table SA.5 of the Supplementary Appendix, we list the matched pairs for directly and indirectly treated bonds.

C Proofs

C.1 Proof of Proposition 1

C.1.1 Cutoffs

Recall that payoffs are piecewise linear in output z with cutoffs defined in Section 4.1. Each is determined by equating pairs of the sovereign's payoffs Π^a in equations (3)–(5):

- Solving $\Pi^\emptyset = \Pi^f$ gives

$$z^{\emptyset \succ f} = \frac{(1 + \omega_f)D_f + \omega_f D_r}{\omega_f + c_f}. \quad (12)$$

- Solving $\Pi^\emptyset = \Pi^{rf}$ gives

$$z^{\emptyset \succ rf} = \frac{(1 + \omega_f + \omega_r)(D_f + D_r)}{1 + \omega_f + \omega_r - (1 - c_f)(1 - c_r)}. \quad (13)$$

- Solving $\Pi^f = \Pi^{rf}$ gives

$$z^{f \succ rf} = \frac{(1 + \omega_f + \omega_r)D_r}{(1 - c_f)(\omega_r + (1 + \omega_f)c_r)}. \quad (14)$$

C.1.2 Derivative of p_r w.r.t. ω_r

We take each regime in turn.

Selective default regime. Here, per equation (7), the price of r debt is

$$p_r = 1 - F\left(z^{f \succ rf}\right) + \int_0^{z^{f \succ rf}} \frac{\omega_r}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z}{D_r} f(z) dz. \quad (15)$$

Differentiating gives

$$\begin{aligned} \frac{\partial p_r}{\partial \omega_r} = & \left\{ -1 + \frac{\omega_r}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z^{f \succ rf}}{D_r} \right\} f\left(z^{f \succ rf}\right) \frac{\partial z^{f \succ rf}}{\partial \omega_r} \\ & + \int_0^{z^{f \succ rf}} \frac{\partial}{\partial \omega_r} \left(\frac{\omega_r}{1 + \omega_f + \omega_r} \right) \frac{(1 - c_f)(1 - c_r)z}{D_r} f(z) dz, \end{aligned} \quad (16)$$

where, from equation (14),

$$\frac{\partial z^{f \succ rf}}{\partial \omega_r} = -\frac{D_r(1 + \omega_f)(1 - c_r)}{(1 - c_f)(\omega_r + (1 + \omega_f)c_r)^2} < 0. \quad (17)$$

The term in braces in equation (16) is strictly negative at the cutoff $z^{f \succ rf}$, which can be seen by plugging in directly from equation (14). Since the integrand in the second term is positive, both terms in equation (16) are positive. Therefore, in this regime,

$$\frac{\partial p_r}{\partial \omega_r} > 0. \quad (18)$$

Cross-default regime. Here, per equation (7), the price of r debt is

$$p_r = 1 - F\left(z^{\mathcal{O} \succ r f}\right) + \int_0^{z^{\mathcal{O} \succ r f}} \frac{\omega_r}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z}{D_r} f(z) dz. \quad (19)$$

Differentiation gives

$$\begin{aligned} \frac{\partial p_r}{\partial \omega_r} = & \left\{ -1 + \frac{\omega_r}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z^{\mathcal{O} \succ r f}}{D_r} \right\} f\left(z^{\mathcal{O} \succ r f}\right) \frac{\partial z^{\mathcal{O} \succ r f}}{\partial \omega_r} \\ & + \int_0^{z^{\mathcal{O} \succ r f}} \frac{\partial}{\partial \omega_r} \left(\frac{\omega_r}{1 + \omega_f + \omega_r} \right) \frac{(1 - c_f)(1 - c_r)z}{D_r} f(z) dz, \end{aligned} \quad (20)$$

where, from equation (13),

$$\frac{\partial z^{\mathcal{O} \succ r f}}{\partial \omega_r} = -\frac{(D_f + D_r)(1 - c_f)(1 - c_r)}{\left((1 + \omega_f + \omega_r) - (1 - c_f)(1 - c_r)\right)^2} < 0. \quad (21)$$

Given that the integrand in equation (20) is positive, the derivative $\partial p_r / \partial \omega_r$ is positive as long as the term in braces is sufficiently negative. That always holds, as can be seen from substituting $z^{\mathcal{O} \succ r f}$ and $\partial z^{\mathcal{O} \succ r f} / \partial \omega_r$ into equation (20).

C.2 Proof of Proposition 2

C.2.1 Derivative of p_f w.r.t. ω_r

We take each regime in turn.

Selective default regime. Here, per equation (7), the price of f debt is

$$\begin{aligned} p_f = & 1 - F\left(z^{\mathcal{O} \succ f}\right) \\ & + \int_{z^f \succ r f}^{z^{\mathcal{O} \succ f}} \frac{\omega_f}{1 + \omega_f} \frac{(1 - c_f)z - D_r}{D_f} f(z) dz \\ & + \int_0^{z^f \succ r f} \frac{\omega_f}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z}{D_f} f(z) dz. \end{aligned} \quad (22)$$

Differentiating gives

$$\begin{aligned} \frac{\partial p_f}{\partial \omega_r} = & \left\{ \frac{\omega_f}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z^{f \succ rf}}{D_f} - \frac{\omega_f}{1 + \omega_f} \frac{(1 - c_f)z^{f \succ rf} - D_r}{D_f} \right\} f(z^{f \succ rf}) \frac{\partial z^{f \succ rf}}{\partial \omega_r} \\ & - \int_0^{z^{f \succ rf}} \frac{\omega_f}{(1 + \omega_f + \omega_r)^2} \frac{(1 - c_f)(1 - c_r)z}{D_f} f(z) dz. \end{aligned} \quad (23)$$

The term in braces is zero at the cutoff $z^{f \succ rf}$, which you can see either by plugging in directly from equation (14) or by realizing that the payoff is continuous at $z^{f \succ rf}$, leading the boundary terms to cancel. The derivative becomes

$$\frac{\partial p_f}{\partial \omega_r} = - \int_0^{z^{f \succ rf}} \frac{\omega_f}{(1 + \omega_f + \omega_r)^2} \frac{(1 - c_f)(1 - c_r)z}{D_f} f(z) dz, \quad (24)$$

which is always negative.

Cross-default regime. Here, per equation (7), the price of f debt is

$$p_f = 1 - F(z^{\emptyset \succ rf}) + \int_0^{z^{\emptyset \succ rf}} \frac{\omega_f}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z}{D_f} f(z) dz. \quad (25)$$

Differentiation gives

$$\begin{aligned} \frac{\partial p_f}{\partial \omega_r} = & \left\{ -1 + \frac{\omega_f}{1 + \omega_f + \omega_r} \frac{(1 - c_f)(1 - c_r)z^{\emptyset \succ rf}}{D_f} \right\} f(z^{\emptyset \succ rf}) \frac{\partial z^{\emptyset \succ rf}}{\partial \omega_r} \\ & - \int_0^{z^{\emptyset \succ rf}} \frac{\omega_f}{(1 + \omega_f + \omega_r)^2} \frac{(1 - c_f)(1 - c_r)z}{D_f} f(z) dz, \end{aligned} \quad (26)$$

where, from equation (13),

$$\frac{\partial z^{\emptyset \succ rf}}{\partial \omega_r} = - \frac{(D_f + D_r)(1 - c_f)(1 - c_r)}{\left((1 + \omega_f + \omega_r) - (1 - c_f)(1 - c_r) \right)^2} < 0. \quad (27)$$

Given the integrand in equation (26) is positive, the derivative $\partial p_f / \partial \omega_r$ is positive whenever the term in braces is sufficiently negative. Substituting $z^{\emptyset \succ rf}$ and $\partial z^{\emptyset \succ rf} / \partial \omega_r$ into equation (26) gives the necessary and sufficient condition for that to be the case, as follows.

The term in braces is negative if

$$\omega_f(1 - c_f)(1 - c_r)(D_r + D_f) < D_f[1 + \omega_f + \omega_r - (1 - c_f)(1 - c_r)]. \quad (28)$$

Under this condition, the crime watch effect of an increase in ω_r from deterring default dominates the multilateral bargaining gain to rigid from flexible creditors.

Substituting $z^{\varnothing \succ r f}$ from equation (13) and $\partial z^{\varnothing \succ r f} / \partial \omega_r$ from equation (27) into equation (26), and using (28), $\partial p_f / \partial \omega_r$ is positive if and only if

$$\left(1 - \frac{\omega_f(1 - c_f)(1 - c_r)(D_f + D_r)}{D_f(1 + \omega_r + \omega_f - (1 - c_r)(1 - c_f))}\right) f(z^{\varnothing \succ r f}) > \frac{\omega_f(D_f + D_r)}{D_f(z^{\varnothing \succ r f})^2} \int_0^{z^{\varnothing \succ r f}} z f(z) dz. \quad (29)$$

Since $\int_0^{z^{\varnothing \succ r f}} z f(z) dz = F(z^{\varnothing \succ r f})\bar{z}$ where $\bar{z} := \mathbb{E}^Q[z \mid z < z^{\varnothing \succ r f}]$, this can be written as

$$\frac{f(z^{\varnothing \succ r f})}{F(z^{\varnothing \succ r f})} > \frac{\omega_f(D_f + D_r)}{D_f(z^{\varnothing \succ r f})^2} \frac{\bar{z}}{1 - \frac{\omega_f(1 - c_f)(1 - c_r)(D_f + D_r)}{D_f(1 + \omega_r + \omega_f - (1 - c_r)(1 - c_f))}}. \quad (30)$$

This condition says that the crime watch spillover outweighs the burglar alarm. It holds if $f(z^{\varnothing \succ r f})/F(z^{\varnothing \succ r f})$ is sufficiently large relative to $\bar{z}/(z^{\varnothing \succ r f})^2$ —in particular, it holds when the default probability $F(z^{\varnothing \succ r f})$ is relatively small, consistent with the empirical observations in our sample.

C.3 Proof of Proposition 3

To derive the result, we use the definition of yield, a linear approximation, and the chain rule:

1. By definition, the spread on a (zero-coupon) bond of type $i \in \{r, f\}$ is $s_i = -\frac{1}{T_i} \log p_i$ (keeping in mind our notational convention that the price is inflated by the risk-free rate).
2. Differentiating the price, we get

$$dp_i = de^{-s_i T_i} = -e^{-s_i T_i} T_i ds_i = -p_i T_i ds_i. \quad (31)$$

3. Rearranging, approximating, and applying the chain rule, we get

$$\Delta s_i \approx -\frac{\Delta p_i}{p_i T_i} \approx -\frac{1}{p_i T_i} \frac{\partial p_i}{\partial h_r} \Delta h_r. \quad (32)$$

Using derivatives of equation (7) with respect to h_r and substitute for $\frac{\partial p_i}{\partial h_r}$, we have the system in the statement of the proposition. $\square$

C.4 Proof of Proposition 4

The elasticity of the default probability with respect to the average haircut is

$$\mathcal{E} \equiv \frac{dF(z^*)/F(z^*)}{d\bar{h}/\bar{h}} = \frac{\bar{h}}{F(z^*)} f(z^*) \frac{dz^*}{d\bar{h}}. \quad (33)$$

We work in the cross-default regime where the relevant threshold $z^* = z^{\varnothing \succ r f}$. From equation (21):

$$\frac{dz^*}{d\omega_r} = -\frac{(D_r + D_f)(1 - c_r)(1 - c_f)}{(1 + \omega_r + \omega_f - (1 - c_r)(1 - c_f))^2}. \quad (34)$$

From the proof of Proposition 3, differentiating $\bar{h} = \varphi h_r + (1 - \varphi)h_f$ with respect to ω_r :

$$\frac{d\bar{h}}{d\omega_r} = \varphi \frac{dh_r}{d\omega_r} + (1 - \varphi) \frac{dh_f}{d\omega_r} = -\frac{\varphi(1 - h_r)(1 + \omega_f)}{\omega_r(1 + \omega_r + \omega_f)} + \frac{(1 - \varphi)(1 - h_f)}{1 + \omega_r + \omega_f}. \quad (35)$$

Using $1 - h_r = \omega_r \mu / \varphi$ and $1 - h_f = \omega_f \mu / (1 - \varphi)$, where $\mu := (1 - c_r)(1 - c_f)\bar{z} / ((1 + \omega_r + \omega_f)D)$ and $\bar{z} := \mathbb{E}^{\mathbb{Q}}[z \mid z < z^*]$:

$$\begin{aligned} \frac{d\bar{h}}{d\omega_r} &= \frac{1}{1 + \omega_r + \omega_f} \left[-\frac{\varphi(1 + \omega_f)}{\omega_r} \cdot \frac{\omega_r \mu}{\varphi} + (1 - \varphi) \cdot \frac{\omega_f \mu}{1 - \varphi} \right] \\ &= \frac{\mu}{1 + \omega_r + \omega_f} [\omega_f - (1 + \omega_f)] = -\frac{\mu}{1 + \omega_r + \omega_f}. \end{aligned} \quad (36)$$

Using (34) and (36), we can write

$$\frac{dz^*}{d\bar{h}} = \frac{D(1-c_r)(1-c_f)}{(1+\omega_r+\omega_f-(1-c_r)(1-c_f))^2} \cdot \frac{1+\omega_r+\omega_f}{\mu}. \quad (37)$$

From the proof of Proposition 3:

$$f(z^*) = F_{h_r} \cdot \frac{(1+\omega_r+\omega_f-(1-c_r)(1-c_f))^2(1-h_r)(1+\omega_f)}{D(1-c_r)(1-c_f)\omega_r(1+\omega_r+\omega_f)}. \quad (38)$$

Multiplying (38) by $dz^*/d\bar{h}$ from (37) and simplifying:

$$f(z^*)\frac{dz^*}{d\bar{h}} = \frac{F_{h_r}(1-h_r)(1+\omega_f)}{\omega_r\mu}. \quad (39)$$

Finally, substituting $\mu = (1-\bar{h})/(\omega_r+\omega_f)$ we obtain:

$$f(z^*)\frac{dz^*}{d\bar{h}} = \frac{F_{h_r}(1-h_r)(1+\omega_f)(\omega_r+\omega_f)}{\omega_r(1-\bar{h})}. \quad (40)$$

Substituting (40) into (33) completes the proof.

D Details of Quantitative Analysis in subsections 4.3 and 4.4

Here we describe omitted details from our quantitative analysis.

D.1 Data Selection

We begin with the full cross-section of our sample of bonds the days before our baseline experiments (May 29, 2012 and November 23, 2012, respectively) and we retain only bonds that are (i) issued under English, New York, or local law, (ii) not missing yield or credit spread data, and (iii) issued in US dollars, euros, or local currency.³⁶ For the English-law experiment, we are left with sixteen countries for which we observe both local- and English-law bonds that

³⁶We discard the few bonds issued in Japanese yen, Swiss francs, or British pounds to simplify the credit spread currency adjustment below.

satisfy all criteria; for the New York-law experiment, we are left with twelve countries.³⁷

D.2 Yield Adjustments

We need to adjust the raw yields to account for our modeling assumptions, namely that risk-free rates are normalized to zero and that all bonds are denominated in the same currency:

- Risk-free rate: To undo the rate normalization, we can just replace yields in the model with credit spreads in the data (this follows immediately from multiplying both sides of equation (7) by e^{-r_f}).
- Currency risk: The model is written from the point of view of a US investor, that is, under the dollar risk-neutral measure.³⁸ We therefore need to adjust observed credit spreads on non-dollar bonds (see the “quanto adjustment” in Du and Schreger (2016)). The intuition is as follows: if the dollar is expected to appreciate against the currency of denomination (say, the euro) in the event of default, the dollar-denominated credit spread will exceed the observed euro-denominated credit spread even if the default probability and percentage haircut are equal for dollar- and euro-bonds. This is because, expressed in a common numéraire, the dollar haircut exceeds the euro haircut. As a result, dollar-denominated CDS on Eurozone sovereigns tend to have higher premiums than euro-denominated ones, even though they share a common trigger event and recovery. Since euro-denominated CDS quotes tend to be reliable for most sovereigns, we adjust the credit spread on euro-denominated bonds by adding the difference between US dollar and euro denominated CDS premiums. For emerging markets sovereigns, however, local-currency CDS quotes are often unreliable (Du and Schreger, 2016). We therefore use the dollar-denominated five-year CDS premium as a proxy for the currency-adjusted credit spread in local-currency, local-law bonds.³⁹

³⁷Those countries are AT, BE, CZ, DK, ES, FI, HR, HU, MY, PH, PL, RO, RU, SE, SK, and TR for English law. For New York law, they are BG, CL, CO, CR, FI, HU, IT, PH, PL, RO, TR, ZA.

³⁸I.e. the Q measure in the model should be interpreted as the one that makes asset prices discounted by the US dollar-denominated money market account martingales.

³⁹While sovereign CDS in emerging markets are not triggered by default on local-currency, local-law bonds, using

For each country, we define the spread variable s_i as the average currency-adjusted credit spread on its English-/local-law bonds for $i = r/f$.

D.3 Variables Definitions and Values

We use the following variables to solve the system in Proposition 3 and to compute the elasticity $\mathcal{E}$ in Proposition 4.

- Credit spreads: $s_r = 2.52\%$ and $s_f = 2.47\%$ from equal-weighted average of (currency-adjusted) English- and local-law spreads across countries. For the New York-law experiment, these numbers are $s_r = 1.88\%$ and $s_f = 1.76\%$ for (currency-adjusted) New York- and local-law spreads.
- Maturity: $T_r = T_f = 5$ from assuming a flat term structure to match the maturity of the most reliable CDS quotes. (For most countries in the English-law experiment, bonds in this sample have a similar average maturity of about six years anyway).
- Q-default probability: $F(z^*) = 1 - e^{-\frac{\xi T}{1-R}} = 17.5\%$ for the English-law experiment and 10.8% for the New York-law experiment, per the “triangle method” (Hébert and Schreger (2017, Appendix A.1)). Here ξ is the T -year CDS premium and R the recovery value from the Markit CDS data. We use $T = 5$ years because the five-year premiums are deemed most reliable (Hébert and Schreger (2017)). We compute the five-year default probability for each of the countries involved in this exercise and take an equal-weighted average.
- Prices p_i : $p_i = e^{-s_i T_i}$ by definition.
- Haircuts: $h_i = (1 - p_i)/F(z^*)$ from the model (equation (7)).
- Fraction of English-law bonds: $\varphi = 20.5\%$ (equal-weighted cross-country average.)
- Fraction of NY-law bonds: $\varphi = 14.6\%$ (equal-weighted cross-country average.)
- Average haircut: $\bar{h} = \varphi h_r + (1 - \varphi) h_f$ by definition.

the CDS premium as a proxy for currency adjusted credit spreads is consistent with our model, in which the default event coincides between local- and foreign-law bonds.

- Spread changes: $\Delta s_r = 5.80$ bps and $\Delta s_f = 4.25$ bps, and, respectively, $\Delta s_r = 33.88$ bps and $\Delta s_f = 16.03$ bps, from Table 3 (these are the estimates from the matched regression, which is more representative of the restricted sample used here, of bonds issued by countries with both directly and indirectly treated bonds outstanding).
- The sensitivity of default to h_r : $F_{h_r} = 0.89$, from solving the system in Proposition 3 for the English-law experiment, and $F_{h_r} = 0.17$ for the New York-law experiment.

D.4 Variable Values for Argentina Policy Counterfactual

For the policy counterfactual in Section 4.4, we use data from Argentina the day before our ruling to be consistent with the other variables. The default costs, taken from Hébert and Schreger (2017), are estimated over a longer period that includes that date.

- Q-default probability: $F(z^*) = 87\%$ applying the triangle method described above for five-year dollar-denominated CDS on Argentina.
- Haircut: $\bar{h} = 66\%$ (based on $F(z^*)$, $D_r = 31.6\%$ —calculated as the fraction of NY-law and English-law debt relative to total debt—and Argentine credit spreads on dollar-denominated five-year bonds under local and New York law. For this exercise, we are careful to derive credit spreads from bonds that are not in default at the time of our baseline ruling).
- Total default cost: $cz^* = 27.8$ billion USD from Hébert and Schreger (2017, Table 3). They describe this number as the cost of the Q-default probability going from zero to one, which arguably corresponds to the cost at our threshold z^* .

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SOVEREIGN BOND RESTRUCTURING: SUPPLEMENTARY APPENDIX

April 24, 2026

Figure SA.1: **Pre- and post-event trends.** We plot the dynamic development of bond yields net of risk-free rates around the English-law experiment (the High Court’s ruling in *Azevedo v Imcopa* on May 30th, 2012). In line with the baseline results, point estimates and standard errors are computed from regressions of 2-day yield *changes* around the event but plotted in terms of yield levels for ease of interpretation. The figure reflects regression results from the baseline specification with rating fixed effects. As in that baseline, standard errors are clustered by country and reported as 90% confidence intervals.

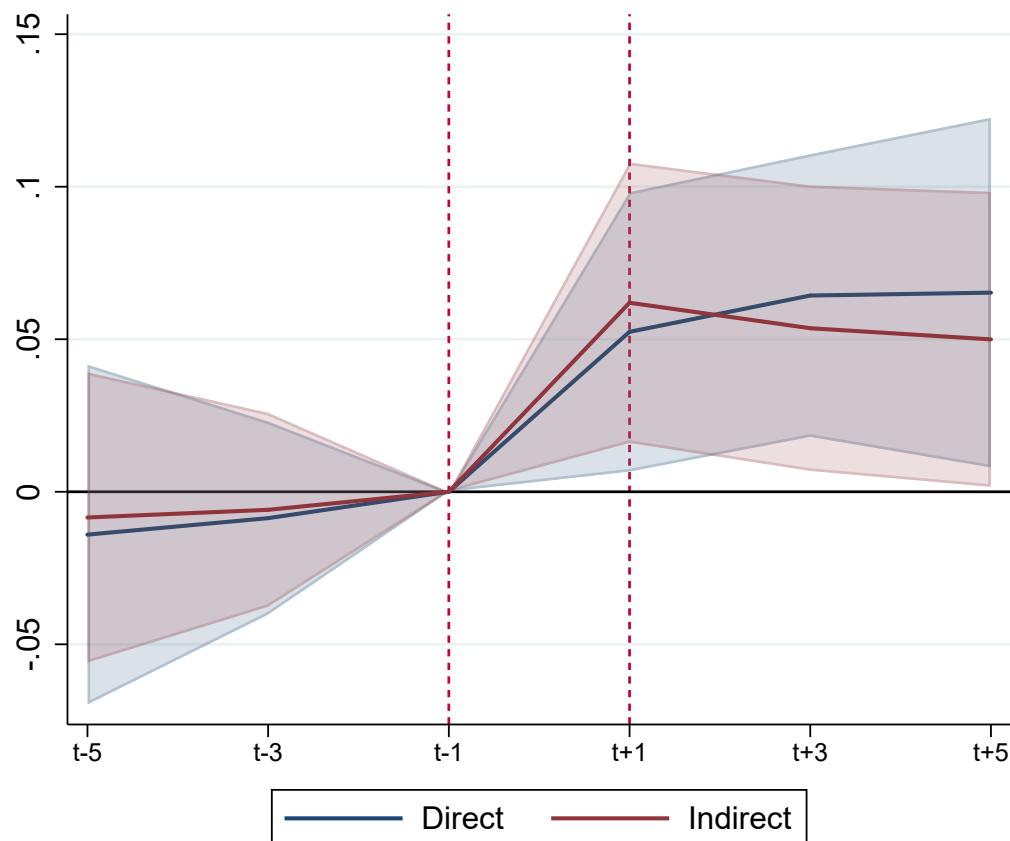


Table SA.1: **Swap rates used in credit spread adjustment.** We list the different currencies in our sample along with the floating rate underlying the swap used to construct credit spreads (and the Bloomberg ticker for the swap rate).

Currency	Swap rate
AUD Australian dollar	Swap vs 6-month Bank Bill rate (ADSW)
BGN Bulgarian lev	Swap rate unavailable
BRL Brazilian real	Swap vs one-day Brazil Cetip Interbank Deposit rate (BCSW)
CAD Canadian dollar	Swap vs 3-month CDOR (CDSW)

Continued on next page

Swap rates (continued)

Currency	Swap rate
CHF Swiss franc	Swap vs SARON (SFSNT)
CLP Chilean peso	Swap vs interbank overnight rate CAMARA (CHSWP)
COP Colombian peso	Swap rate unavailable
CNY Chinese renminbi	Swap vs CNY 7-day repo rate (CCSWO)
CZK Czech koruna	Swap vs 6-month Prague IBOR (CKSW)
DKK Danish krone	Swap vs 6-month Copenhagen IBOR (DKSW)
DOP Dominican peso	Swap rate unavailable
EGP Egyptian pound	Swap rate unavailable
EUR euro	Swap vs EONIA (EUSWE)
GBP British pound	Swap vs SONIA (BPSW)
GHS Ghana cedi	Swap rate unavailable
HRK Croatian kuna	Swap rate unavailable
HUF Hungarian forint	Swap vs 6-month Budapest IBOR (HFSW)
IDR Indonesian rupiah	Swap rate unavailable
ISK Icelandic krone	Swap rate unavailable
JPY Japanese yen	Swap vs 6-month ICE JPY LIBOR (JYSW)
MXN Mexican peso	Swap vs 28-day Interbank TIE (MPSW)
MYR Malaysian ringgit	Swap vs 3-month Kuala Lumpur IBOR (MRSWQO)
NGN Nigerian naira	Swap rate unavailable
NOK Norwegian krone	Swap vs 6-month Norway IBOR (NKSX)
PEN Peruvian sol	Swap rate unavailable
PHP Philippines peso	Swap vs 3-month Philippine Interbank Reference Rate (PPSWO)
PLN Polish zloty	Swap vs 6-month Warsaw IBOR (PZSW)
RON Romanian leu	Swap vs 3-month Bucharest Interbank Offer/Bid Rates (RNSW)
RUB Russian ruble	Swap vs 3-month Moscow Prime Offered Rate (RRSWM)
SEK Swedish krona	Swap vs 3-month Stockholm IBOR (SKSW)
SGD Singapore dollar	Swap vs 6-month Association of Banks in Singapore Swap Offer Rate (SDSW)
TRY Turkish lira	Swap vs TRY Overnight Reference rate (TYSO)
UAH Ukrainian Hryvnia	Swap rate unavailable
USD US dollar	Swap vs US Federal Funds Effective Rate (USSO)
UYU Uruguayan peso	Swap rate unavailable
ZAR South African rand	Swap vs 3-month Johannesburg Interbank Agreed Rate (SASW)

Table SA.2: **Matched pairs, English-law experiment: Directly treated bonds.** We list 124 directly treated bond pairs below with bond characteristics for treated (English law, left) and control (New York law, right) bonds in each column. Yields and credit spreads are as of the day before the baseline ruling (May 29, 2012).

Country	Currency	Maturity	Yield	Credit spread	Rating
AE	CO	USD	10/5/2015	2/15/2016	4.53 4.24 4.23 3.76 A BBB+
AE	MX	USD	11/3/2014	3/3/2015	3.97 1.52 3.76 1.22 A BBB+
AE	MX	USD	4/8/2019	3/19/2019	2.74 2.82 2.07 2.14 A BBB+
AE	MX	USD	4/8/2014	1/15/2014	1.04 1.11 0.83 0.90 A BBB+
AE	MX	USD	4/8/2014	1/15/2014	1.05 1.11 0.84 0.90 A BBB+
AE	MX	USD	7/22/2014	3/3/2015	2.60 1.52 2.40 1.22 A BBB+
AE	MX	USD	4/8/2019	3/19/2019	2.74 2.82 2.06 2.14 A BBB+
AE	ZA	USD	10/5/2020	3/9/2020	6.27 3.77 3.43 0.93 A BBB+
AE	ZA	USD	6/22/2021	5/30/2022	5.16 4.05 2.32 1.21 A BBB+
AE	ZA	USD	5/2/2022	5/30/2022	6.13 4.05 3.29 1.21 A BBB+
AE	ZA	USD	5/2/2017	6/23/2017	4.46 3.55 3.78 2.87 A BBB+
BB	PA	USD	8/4/2022	1/16/2023	6.74 5.03 3.90 2.18 BBB BBB
BB	UY	USD	12/5/2035	3/21/2036	7.53 4.94 5.36 2.78 BBB BB+
BB	ZA	USD	12/15/2021	5/30/2022	6.81 4.05 3.97 1.21 BBB BBB+
BH	CO	USD	6/30/2015	2/15/2016	4.79 4.24 4.49 3.76 BBB BBB+
BH	MX	USD	3/18/2013	2/17/2014	2.98 1.18 2.81 0.98 BBB BBB+
BH	PA	USD	6/17/2014	3/15/2015	2.80 1.68 2.59 1.38 BBB BBB
BH	ZA	USD	11/22/2018	5/27/2019	4.97 3.57 4.29 2.89 BBB BBB+
BH	ZA	USD	3/31/2020	5/27/2019	5.78 3.57 2.94 2.89 BBB BBB+
BH	ZA	USD	3/31/2020	5/27/2019	5.79 3.57 2.94 2.89 BBB BBB+
BR	MX	EUR	6/26/2017	7/14/2017	2.50 2.55 1.81 1.86 BBB BBB+
BY	DO	USD	1/26/2018	1/23/2018	11.11 5.39 10.43 4.71 B B
BY	LK	USD	8/3/2015	1/22/2015	10.80 5.29 10.50 4.99 B B+
ES	MX	USD	5/9/2013	2/17/2014	3.01 1.18 2.83 0.98 BBB+ BBB+
ES	MX	USD	5/9/2013	2/17/2014	2.46 1.18 2.29 0.98 BBB+ BBB+
ES	PA	USD	5/5/2014	3/15/2015	4.44 1.68 4.23 1.38 BBB+ BBB
GE	LK	USD	4/12/2021	10/4/2020	5.82 6.40 2.97 3.55 B+ B+
GE	LK	USD	4/12/2021	10/4/2020	5.85 6.40 3.01 3.55 B+ B+
HR	PA	USD	12/3/2022	1/16/2023	7.53 5.03 4.69 2.18 BBB− BBB
HR	PA	USD	12/3/2022	1/16/2023	7.52 5.03 4.68 2.18 BBB− BBB
HR	PE	EUR	4/15/2014	10/14/2014	2.88 1.82 2.59 1.53 BBB− BBB−
HR	UY	EUR	9/7/2018	6/28/2019	6.32 4.30 5.64 3.61 BBB− BB+
HR	UY	EUR	5/6/2015	1/19/2016	4.68 2.38 4.30 1.86 BBB− BB+
HR	ZA	USD	5/11/2019	5/27/2019	7.49 3.57 6.81 2.89 BBB− BBB+
HR	ZA	USD	2/7/2021	3/9/2020	7.50 3.77 4.66 0.93 BBB− BBB+
HR	ZA	USD	2/7/2021	3/9/2020	7.57 3.77 4.72 0.93 BBB− BBB+

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
HR	ZA	USD	3/4/2019	5/27/2019	7.04	3.57	6.36	2.89	BBB−	BBB+
HR	ZA	USD	5/11/2019	5/27/2019	7.50	3.57	6.82	2.89	BBB−	BBB+
HR	ZA	USD	3/4/2019	5/27/2019	7.01	3.57	6.33	2.89	BBB−	BBB+
HU	MX	EUR	6/1/2017	5/8/2017	7.75	2.63	7.06	1.94	BBB−	BBB+
HU	MX	EUR	4/2/2017	5/8/2017	8.00	2.63	7.31	1.94	BBB−	BBB+
HU	MX	EUR	2/6/2013	4/8/2013	5.59	1.28	5.34	1.03	BBB−	BBB+
HU	PE	EUR	1/29/2014	10/14/2014	7.05	1.82	6.76	1.53	BBB−	BBB−
HU	UY	EUR	4/7/2016	1/19/2016	7.09	2.38	6.57	1.86	BBB−	BB+
HU	UY	EUR	11/6/2018	6/28/2019	8.27	4.30	7.58	3.61	BBB−	BB+
HU	UY	EUR	11/5/2019	6/28/2019	8.33	4.30	7.64	3.61	BBB−	BB+
ID	PA	USD	4/23/2014	3/15/2015	2.87	1.68	2.66	1.38	BB+	BBB
ID	PA	USD	4/23/2014	3/15/2015	2.73	1.68	2.53	1.38	BB+	BBB
ID	PE	USD	11/21/2018	3/30/2019	4.08	2.74	3.40	2.06	BB+	BBB−
ID	PE	USD	11/21/2018	3/30/2019	4.09	2.74	3.41	2.06	BB+	BBB−
IS	ZA	USD	4/6/2017	6/23/2017	4.82	3.55	4.14	2.87	BBB−	BBB+
JO	LK	USD	11/12/2015	1/22/2015	4.38	5.29	4.08	4.99	BB	B+
LB	SV	USD	3/9/2020	12/1/2019	5.74	5.79	2.89	2.95	B	BB−
LT	CO	USD	3/10/2016	2/15/2016	3.75	4.24	3.28	3.76	BBB	BBB+
LT	CO	USD	3/10/2016	2/15/2016	3.80	4.24	3.32	3.76	BBB	BBB+
LT	MX	EUR	3/5/2013	4/8/2013	1.49	1.28	1.24	1.03	BBB	BBB+
LT	UY	EUR	7/10/2018	6/28/2019	4.63	4.30	3.94	3.61	BBB	BB+
LT	UY	EUR	10/6/2015	1/19/2016	2.44	2.38	2.07	1.86	BBB	BB+
LT	UY	EUR	7/4/2018	6/28/2019	4.61	4.30	3.92	3.61	BBB	BB+
LT	UY	EUR	2/10/2016	1/19/2016	3.41	2.38	2.89	1.86	BBB	BB+
LT	ZA	USD	2/9/2018	6/23/2017	4.44	3.55	3.76	2.87	BBB	BBB+
LT	ZA	USD	1/2/2022	5/30/2022	5.50	4.05	2.65	1.21	BBB	BBB+
LT	ZA	USD	1/2/2022	5/30/2022	5.45	4.05	2.60	1.21	BBB	BBB+
LT	ZA	USD	9/3/2021	5/30/2022	5.30	4.05	2.45	1.21	BBB	BBB+
LT	ZA	USD	11/2/2020	3/9/2020	5.15	3.77	2.31	0.93	BBB	BBB+
LT	ZA	USD	9/3/2021	5/30/2022	5.33	4.05	2.49	1.21	BBB	BBB+
LT	ZA	USD	11/2/2020	3/9/2020	5.14	3.77	2.30	0.93	BBB	BBB+
LT	ZA	USD	2/9/2018	6/23/2017	4.44	3.55	3.76	2.87	BBB	BBB+
LV	MX	EUR	5/3/2018	5/8/2017	3.78	2.63	3.10	1.94	BBB	BBB+
LV	PA	USD	4/6/2022	1/16/2023	5.72	5.03	2.88	2.18	BBB	BBB
LV	PE	EUR	4/2/2014	10/14/2014	2.06	1.82	1.78	1.53	BBB	BBB−
LV	ZA	USD	10/2/2018	5/27/2019	5.12	3.57	4.44	2.89	BBB	BBB+
LV	ZA	USD	10/2/2018	5/27/2019	5.14	3.57	4.46	2.89	BBB	BBB+
ME	UY	EUR	8/4/2016	1/19/2016	9.09	2.38	8.57	1.86	BB	BB+
ME	UY	EUR	2/9/2016	1/19/2016	9.19	2.38	8.67	1.86	BB	BB+
MK	UY	EUR	8/12/2015	1/19/2016	6.25	2.38	5.88	1.86	BB	BB+

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
MY	MX	USD	7/6/2016	1/15/2017	2.17	2.34	1.70	1.66	A−	BBB+
MY	MX	USD	7/6/2016	1/15/2017	2.17	2.34	1.69	1.66	A−	BBB+
MY	MX	USD	7/6/2021	3/15/2022	3.28	3.31	0.44	0.46	A−	BBB+
MY	PA	USD	6/4/2015	3/15/2015	1.90	1.68	1.61	1.38	A−	BBB
MY	PA	USD	6/4/2015	3/15/2015	1.79	1.68	1.51	1.38	A−	BBB
NG	LK	USD	1/28/2021	10/4/2020	5.55	6.40	2.71	3.55	B+	B+
PH	UY	USD	10/7/2016	5/17/2017	2.47	2.28	2.00	1.60	BB	BB+
PL	MX	EUR	3/29/2017	5/8/2017	2.82	2.63	2.13	1.94	A−	BBB+
PL	MX	EUR	8/6/2019	2/17/2020	3.30	3.33	2.61	1.96	A−	BBB+
PL	MX	EUR	3/10/2020	2/17/2020	3.52	3.33	2.14	1.96	A−	BBB+
PL	MX	EUR	2/5/2013	6/10/2013	0.97	1.17	0.72	0.92	A−	BBB+
PL	MX	EUR	3/2/2014	4/8/2013	1.65	1.28	1.36	1.03	A−	BBB+
PL	ZA	EUR	1/2/2016	4/5/2016	2.44	2.37	1.92	1.85	A−	BBB+
RO	SV	USD	7/2/2022	1/24/2023	6.64	6.43	3.80	3.59	BB+	BB−
RO	UY	EUR	6/3/2016	1/19/2016	4.42	2.38	3.90	1.86	BB+	BB+
RS	SV	USD	4/9/2023	1/24/2023	7.00	6.43	4.16	3.59	BB	BB−
RS	SV	USD	1/4/2024	1/24/2023	7.28	6.43	4.44	3.59	BB	BB−
RU	CO	USD	4/29/2015	11/16/2015	2.27	2.16	1.97	1.86	BBB	BBB+
RU	CO	USD	4/29/2015	11/16/2015	2.27	2.16	1.97	1.86	BBB	BBB+
RU	PA	USD	3/31/2030	4/1/2029	4.15	4.27	1.99	2.29	BBB	BBB
RU	PA	USD	3/31/2030	4/1/2029	4.17	4.27	2.01	2.29	BBB	BBB
RU	UY	USD	6/26/2028	7/15/2027	5.45	4.41	3.47	2.44	BBB	BB+
RU	ZA	USD	4/29/2020	3/9/2020	3.98	3.77	1.14	0.93	BBB	BBB+
RU	ZA	USD	4/4/2017	6/23/2017	3.15	3.55	2.47	2.87	BBB	BBB+
RU	ZA	USD	4/29/2020	3/9/2020	3.94	3.77	1.10	0.93	BBB	BBB+
RU	ZA	USD	4/4/2022	5/30/2022	4.23	4.05	1.39	1.21	BBB	BBB+
RU	ZA	USD	4/4/2017	6/23/2017	3.16	3.55	2.48	2.87	BBB	BBB+
RU	ZA	USD	4/4/2022	5/30/2022	4.23	4.05	1.39	1.21	BBB	BBB+
SK	MX	EUR	5/20/2014	6/10/2013	1.51	1.17	1.22	0.92	A	BBB+
SK	MX	EUR	3/5/2018	5/8/2017	2.78	2.63	2.09	1.94	A	BBB+
SK	ZA	USD	9/5/2023	1/17/2024	4.89	4.38	2.04	1.54	A	BBB+
SK	ZA	USD	9/5/2023	1/17/2024	4.91	4.38	2.06	1.54	A	BBB+
SK	ZA	EUR	9/5/2016	4/5/2016	1.74	2.37	1.22	1.85	A	BBB+
TR	UY	EUR	5/18/2020	6/28/2019	4.81	4.30	3.43	3.61	BB	BB+
TR	UY	EUR	3/1/2016	1/19/2016	3.66	2.38	3.14	1.86	BB	BB+
TR	UY	EUR	4/2/2019	6/28/2019	4.63	4.30	3.94	3.61	BB	BB+
UA	DO	USD	6/11/2013	1/23/2013	9.63	1.28	9.46	1.10	B+	B
UA	DO	USD	5/6/2017	1/23/2018	9.78	5.39	9.20	4.71	B+	B
UA	DO	USD	5/6/2017	1/23/2018	9.81	5.39	9.23	4.71	B+	B
UA	DO	USD	6/11/2013	1/23/2013	9.90	1.28	9.73	1.10	B+	B

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
UA	DO	USD	9/11/2017	1/23/2018	9.84	5.39	9.25	4.71	B+	B
UA	DO	USD	9/11/2017	1/23/2018	9.83	5.39	9.24	4.71	B+	B
UA	DO	USD	2/11/2018	1/23/2018	9.58	5.39	8.91	4.71	B+	B
UA	DO	USD	11/9/2021	5/6/2021	10.09	6.61	7.24	3.77	B+	B
UA	DO	USD	2/11/2018	1/23/2018	9.61	5.39	8.99	4.71	B+	B
UA	DO	USD	11/9/2021	5/6/2021	9.82	6.61	6.98	3.77	B+	B
UA	LK	USD	2/6/2014	1/22/2015	6.36	5.29	6.17	4.99	B+	B+
UA	SV	USD	11/2/2022	1/24/2023	10.28	6.43	7.44	3.59	B+	BB–

Table SA.3: **Matched pairs, English-law experiment: Indirectly treated bonds.** We list 83 indirectly treated bond pairs below with bond characteristics for treated (local-law/English-law issuer, left) and control (local-law/English-law non-issuer, right) bonds in each column. Yields and credit spreads are as of the day before the baseline ruling (May 29, 2012).

Country		Currency	Maturity		Yield		Credit spread		Rating	
AT	FR	EUR	1/15/2018	10/25/2017	1.55	1.48	0.87	0.80	AA+	AAA
AT	FR	EUR	1/26/2062	1/3/2062	2.93	3.25	1.21	1.53	AA+	AAA
AT	FR	EUR	9/15/2021	1/2/2022	2.16	2.16	0.78	0.78	AA+	AAA
AT	FR	EUR	4/20/2022	4/25/2022	2.26	2.34	0.88	0.97	AA+	AAA
AT	FR	EUR	10/20/2013	7/12/2013	0.21	0.21	−0.04	−0.04	AA+	AAA
AT	FR	EUR	9/15/2017	10/25/2017	1.43	1.48	0.75	0.80	AA+	AAA
AT	FR	EUR	3/15/2026	10/25/2025	2.60	2.88	0.91	1.19	AA+	AAA
AT	FR	EUR	12/1/2014	7/12/2015	0.81	0.72	0.44	0.35	AA+	AAA
AT	FR	EUR	2/20/2017	7/25/2016	1.25	1.10	0.56	0.58	AA+	AAA
AT	FR	EUR	11/22/2022	4/25/2022	2.35	2.34	0.97	0.97	AA+	AAA
AT	FR	EUR	9/15/2016	7/25/2016	1.09	1.10	0.57	0.58	AA+	AAA
AT	FR	EUR	1/15/2014	4/25/2014	0.36	0.33	0.08	0.04	AA+	AAA
AT	LU	EUR	10/20/2014	12/4/2013	0.42	0.42	0.14	0.14	AA+	AAA
AT	LU	EUR	3/1/2020	5/18/2020	1.83	1.65	0.45	0.28	AA+	AAA
AT	NL	EUR	3/15/2037	3/4/2038	2.85	2.17	1.11	0.45	AA+	AAA
AT	NL	EUR	7/15/2014	7/15/2015	0.38	0.50	0.09	0.13	AA+	AAA
AT	NL	EUR	7/15/2027	1/15/2028	2.67	2.12	0.98	0.42	AA+	AAA
BE	DE	EUR	10/11/2031	1/4/2031	3.32	1.99	1.58	0.25	AA+	AAA
BE	FR	EUR	4/5/2014	9/25/2014	0.43	0.44	0.15	0.16	AA+	AAA
BE	FR	EUR	4/12/2019	10/25/2019	3.04	2.04	2.35	1.35	AA+	AAA
BE	FR	EUR	6/28/2017	4/25/2018	2.14	1.63	1.45	0.94	AA+	AAA
BE	FR	EUR	1/12/2013	7/12/2013	0.22	0.21	−0.04	−0.04	AA+	AAA
BE	FR	EUR	5/28/2021	4/25/2022	3.25	2.51	1.87	1.13	AA+	AAA

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Matched pairs: Indirectly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
BE	FR	EUR	3/28/2019	10/25/2019	2.55	2.04	1.86	1.35	AA+	AAA
BE	FR	EUR	3/28/2015	2/25/2016	1.18	0.94	0.81	0.43	AA+	AAA
BE	FR	EUR	7/7/2013	4/25/2013	0.15	0.14	-0.10	-0.12	AA+	AAA
BE	FR	EUR	8/9/2013	7/12/2013	0.20	0.21	-0.05	-0.04	AA+	AAA
BE	FR	EUR	2/2/2014	4/25/2014	0.32	0.33	0.04	0.04	AA+	AAA
BE	FR	EUR	3/28/2017	10/25/2017	2.01	1.48	1.32	0.80	AA+	AAA
BE	FR	EUR	3/28/2018	10/25/2018	2.32	1.77	1.64	1.08	AA+	AAA
BE	FR	EUR	3/28/2015	2/25/2016	1.21	0.94	0.83	0.43	AA+	AAA
BE	FR	EUR	4/9/2032	10/25/2032	2.48	3.16	0.74	1.42	AA+	AAA
BE	FR	EUR	4/1/2022	4/25/2022	2.89	2.51	1.51	1.13	AA+	AAA
BE	FR	EUR	4/8/2013	4/25/2013	0.14	0.14	-0.11	-0.12	AA+	AAA
BE	FR	EUR	9/28/2014	7/12/2015	0.92	0.72	0.64	0.35	AA+	AAA
BE	FR	EUR	9/12/2041	4/25/2041	3.50	3.41	1.78	1.69	AA+	AAA
BE	FR	EUR	3/28/2014	10/25/2014	0.72	0.46	0.43	0.17	AA+	AAA
BE	FR	EUR	9/28/2017	4/25/2018	2.21	1.63	1.52	0.94	AA+	AAA
BE	FR	EUR	3/28/2035	4/25/2035	3.60	3.18	1.86	1.44	AA+	AAA
BE	FR	EUR	3/28/2041	4/25/2041	3.62	3.41	1.90	1.69	AA+	AAA
BE	FR	EUR	9/28/2022	4/25/2023	3.20	2.56	1.82	1.18	AA+	AAA
BE	FR	EUR	2/15/2016	10/25/2016	1.39	1.16	0.87	0.64	AA+	AAA
BE	FR	EUR	4/3/2020	10/25/2019	3.23	2.04	1.86	1.35	AA+	AAA
BE	FR	EUR	12/24/2012	10/25/2013	0.25	0.25	0.00	0.00	AA+	AAA
BE	FR	EUR	9/28/2021	4/25/2022	3.04	2.51	1.66	1.13	AA+	AAA
BE	FR	EUR	3/11/2013	7/12/2013	0.21	0.21	-0.04	-0.04	AA+	AAA
BE	FR	EUR	3/28/2016	10/25/2016	1.57	1.16	1.05	0.64	AA+	AAA
BE	FR	EUR	3/28/2032	10/25/2032	3.62	3.16	1.88	1.42	AA+	AAA
BE	FR	EUR	5/23/2022	4/25/2023	3.40	2.56	2.03	1.18	AA+	AAA
BE	FR	EUR	9/28/2019	10/25/2019	2.73	2.04	2.05	1.35	AA+	AAA
BE	FR	EUR	3/28/2022	4/25/2022	3.07	2.51	1.69	1.13	AA+	AAA
BE	FR	EUR	9/28/2016	4/25/2017	1.80	1.34	1.28	0.65	AA+	AAA
BE	FR	EUR	9/28/2013	9/25/2014	0.58	0.44	0.32	0.16	AA+	AAA
BE	FR	EUR	11/10/2031	10/25/2032	3.30	3.16	1.56	1.42	AA+	AAA
BE	FR	EUR	9/28/2015	7/25/2016	1.36	1.10	0.99	0.58	AA+	AAA
BE	LU	EUR	3/28/2013	12/4/2013	0.41	0.42	0.16	0.14	AA+	AAA
BE	LU	EUR	6/4/2014	12/4/2013	0.41	0.42	0.12	0.14	AA+	AAA
BE	LU	EUR	2/3/2014	12/4/2013	0.38	0.42	0.09	0.14	AA+	AAA
BE	NL	EUR	4/6/2028	1/15/2028	3.40	2.12	1.71	0.42	AA+	AAA
BE	NL	EUR	5/1/2014	7/15/2014	0.28	0.28	-0.01	-0.00	AA+	AAA
BE	NL	EUR	3/28/2028	1/15/2028	3.48	2.12	1.78	0.42	AA+	AAA
FI	DE	EUR	3/3/2015	10/9/2015	0.16	0.14	-0.21	-0.23	AAA	AAA
FI	FR	EUR	4/15/2017	1/7/2017	0.79	0.81	0.10	0.13	AAA	AAA

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Matched pairs: Indirectly treated bonds (continued)

Country			Currency		Maturity		Yield		Credit spread		Rating	
FI	FR	EUR	4/10/2025	10/25/2025	2.02	2.88	0.33	1.19	AAA	AAA		
FI	FR	EUR	4/15/2016	1/2/2017	0.47	0.63	−0.05	−0.06	AAA	AAA		
FI	FR	EUR	4/4/2019	1/2/2019	1.33	1.32	0.65	0.63	AAA	AAA		
FI	FR	EUR	7/4/2013	4/5/2013	0.08	0.07	−0.17	−0.18	AAA	AAA		
FI	NL	EUR	7/4/2028	1/15/2028	2.18	2.12	0.48	0.42	AAA	AAA		
FI	NL	EUR	4/15/2020	7/15/2020	1.49	1.51	0.11	0.14	AAA	AAA		
FI	NL	EUR	7/4/2015	7/15/2014	0.35	0.28	−0.02	−0.00	AAA	AAA		
FI	NL	EUR	3/5/2018	3/7/2018	0.92	0.98	0.23	0.29	AAA	AAA		
FI	NL	EUR	4/15/2021	7/15/2021	1.64	1.64	0.26	0.26	AAA	AAA		
IT	PA	USD	8/9/2017	6/15/2018	5.09	3.42	4.41	2.74	BBB+	BBB		
SI	FR	EUR	6/3/2028	4/25/2029	5.85	3.12	4.15	1.42	AA	AAA		
SI	FR	EUR	10/3/2019	10/25/2019	4.24	2.04	3.55	1.35	AA	AAA		
SI	FR	EUR	9/9/2024	10/25/2023	5.53	2.65	4.15	1.27	AA	AAA		
SI	FR	EUR	6/2/2019	10/25/2019	4.84	2.04	4.15	1.35	AA	AAA		
SI	FR	EUR	1/10/2022	4/25/2022	5.49	2.51	4.11	1.13	AA	AAA		
SI	FR	EUR	2/4/2014	10/25/2014	2.36	0.46	2.07	0.17	AA	AAA		
SI	FR	EUR	2/1/2022	4/25/2022	5.19	2.51	3.81	1.13	AA	AAA		
SI	FR	EUR	2/17/2016	10/25/2016	3.57	1.16	3.05	0.64	AA	AAA		
SI	FR	EUR	6/1/2022	4/25/2023	5.28	2.56	3.90	1.18	AA	AAA		
SI	FR	EUR	5/3/2016	4/25/2017	2.64	1.34	2.12	0.65	AA	AAA		

Table SA.4: **Matched pairs, New-York law experiment: Directly treated bonds.** We list 141 directly treated bond pairs below with bond characteristics for treated (New York law, left) and control (English law, right) bonds in each column. Yields and credit spreads are as of the first event (November 26, 2012).

Country			Currency		Maturity		Yield		Credit spread		Rating	
BG	MY	USD	1/15/2015	6/4/2015	1.40	1.34	1.23	1.10	BBB	A−		
BG	MY	USD	1/15/2015	6/4/2015	1.36	1.34	1.19	1.10	BBB	A−		
BS	RU	USD	11/20/2029	3/31/2030	5.06	2.85	3.11	0.90	BBB+	BBB		
BS	RU	USD	11/20/2029	3/31/2030	5.06	2.85	3.11	0.90	BBB+	BBB		
DO	GE	USD	5/6/2021	4/12/2021	5.05	4.60	3.74	3.28	B	B+		
DO	UA	USD	1/23/2018	5/6/2017	3.73	6.89	3.18	6.52	B	B+		
DO	UA	USD	1/23/2018	5/6/2017	3.75	6.89	3.20	6.52	B	B+		
EG	BH	USD	4/29/2020	3/31/2020	5.26	4.43	4.71	3.88	BB+	BBB		
EG	BH	USD	4/29/2020	3/31/2020	5.00	4.43	4.45	3.88	BB+	BBB		
HU	BH	USD	3/2/2015	6/30/2015	3.69	3.59	3.53	3.35	BBB−	BBB		
HU	BH	USD	5/1/2022	7/5/2022	4.79	4.98	3.47	3.67	BBB−	BBB		
HU	BH	USD	5/3/2023	7/5/2022	4.73	4.98	3.42	3.67	BBB−	BBB		

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
ID	BB	USD	10/12/2035	12/5/2035	4.31	7.50	2.15	5.34	BB+	BBB
ID	BB	USD	10/12/2035	12/5/2035	4.30	7.50	2.14	5.34	BB+	BBB
ID	BH	USD	5/4/2014	6/17/2014	1.57	1.64	1.43	1.48	BB+	BBB
ID	BH	USD	3/10/2014	6/17/2014	1.62	1.64	1.48	1.48	BB+	BBB
ID	BH	USD	5/4/2014	6/17/2014	1.55	1.64	1.41	1.48	BB+	BBB
ID	BH	USD	4/20/2015	6/17/2014	1.65	1.64	1.49	1.48	BB+	BBB
ID	BH	USD	4/20/2015	6/17/2014	1.67	1.64	1.50	1.48	BB+	BBB
ID	BH	USD	3/10/2014	6/17/2014	1.64	1.64	1.50	1.48	BB+	BBB
ID	LT	USD	1/17/2018	2/9/2018	2.24	2.32	1.69	1.77	BB+	BBB
ID	LT	USD	1/17/2018	2/9/2018	2.24	2.32	1.69	1.77	BB+	BBB
ID	LT	USD	1/15/2016	3/10/2016	1.84	1.84	1.60	1.60	BB+	BBB
ID	LT	USD	1/15/2016	3/10/2016	1.81	1.83	1.58	1.59	BB+	BBB
ID	LT	USD	3/9/2017	2/9/2018	2.10	2.32	1.72	1.77	BB+	BBB
ID	LT	USD	5/5/2021	11/2/2020	2.90	2.93	1.59	1.62	BB+	BBB
ID	LT	USD	5/5/2021	11/2/2020	2.92	2.93	1.61	1.62	BB+	BBB
ID	LT	USD	3/9/2017	2/9/2018	2.08	2.32	1.71	1.77	BB+	BBB
ID	LV	USD	3/4/2019	10/2/2018	2.63	2.63	2.08	2.08	BB+	BBB
ID	LV	USD	3/4/2019	10/2/2018	2.62	2.63	2.08	2.08	BB+	BBB
ID	RU	USD	3/13/2020	4/29/2020	2.73	2.60	2.18	2.05	BB+	BBB
ID	RU	USD	4/25/2022	4/4/2022	2.94	2.98	1.62	1.67	BB+	BBB
ID	RU	USD	4/25/2022	4/4/2022	2.93	2.98	1.62	1.67	BB+	BBB
ID	RU	USD	3/13/2020	4/29/2020	2.74	2.60	2.19	2.05	BB+	BBB
ID	RU	USD	1/17/2042	4/4/2042	4.28	4.32	1.95	1.99	BB+	BBB
ID	RU	USD	1/17/2042	4/4/2042	4.28	4.32	1.95	1.99	BB+	BBB
IL	AE	USD	6/15/2013	4/8/2014	0.63	0.38	0.49	0.23	A+	A
IL	AE	USD	3/1/2014	4/8/2014	0.89	0.38	0.75	0.23	A+	A
IL	AE	USD	3/26/2019	4/8/2019	2.33	2.09	1.78	1.54	A+	A
IL	MY	USD	6/30/2022	7/6/2021	3.05	2.73	1.73	1.42	A+	A–
IL	MY	USD	11/9/2016	7/6/2016	1.47	1.63	1.10	1.26	A+	A–
IS	HR	USD	11/5/2022	12/3/2022	4.59	4.29	3.28	2.98	BBB–	BBB–
IT	AE	USD	12/6/2017	5/2/2017	3.67	3.42	3.12	3.05	BBB+	A
IT	AE	USD	1/1/2018	5/2/2017	3.24	3.42	2.69	3.05	BBB+	A
IT	AE	USD	6/15/2013	4/8/2014	1.07	0.38	0.92	0.23	BBB+	A
IT	AE	USD	9/1/2016	10/5/2015	2.47	2.79	2.10	2.55	BBB+	A
IT	BH	USD	4/9/2014	6/17/2014	1.25	1.64	1.11	1.48	BBB+	BBB
IT	HR	USD	9/27/2023	12/3/2022	5.04	4.29	3.73	2.98	BBB+	BBB–
IT	LT	USD	2/1/2017	3/10/2016	2.43	1.84	2.06	1.60	BBB+	BBB
LB	BY	USD	8/6/2015	8/3/2015	4.04	8.72	3.81	8.48	B	B
LB	GE	USD	4/12/2021	4/12/2021	5.88	4.60	4.57	3.28	B	B+
LB	UA	USD	1/15/2015	4/8/2014	3.71	7.05	3.54	6.91	B	B+

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
LB	UA	USD	11/12/2018	2/11/2018	5.11	7.19	4.57	6.64	B	B+
LB	UA	USD	11/28/2019	12/7/2018	5.46	7.25	4.91	6.70	B	B+
LB	UA	USD	10/4/2022	11/9/2021	5.26	7.17	3.95	5.86	B	B+
LB	UA	USD	5/11/2016	5/6/2017	4.44	6.89	4.20	6.52	B	B+
LB	UA	USD	3/20/2017	5/6/2017	4.77	6.89	4.40	6.52	B	B+
LB	UA	USD	6/20/2013	6/11/2013	1.89	5.88	1.75	5.74	B	B+
LB	UA	USD	5/20/2019	12/7/2018	5.24	7.25	4.69	6.70	B	B+
LB	UA	USD	5/2/2014	6/11/2013	2.63	5.88	2.49	5.74	B	B+
LB	UA	USD	12/3/2024	4/11/2024	6.22	7.67	4.91	6.36	B	B+
LB	UA	USD	12/31/2017	5/6/2017	4.19	6.89	3.64	6.52	B	B+
LB	UA	USD	1/19/2016	11/9/2016	4.29	6.91	4.06	6.53	B	B+
LB	UA	USD	11/2/2016	5/6/2017	4.66	6.89	4.29	6.52	B	B+
LB	UA	USD	10/12/2017	5/6/2017	4.93	6.89	4.38	6.52	B	B+
LB	UA	USD	4/14/2014	6/11/2013	2.01	5.88	1.87	5.74	B	B+
LK	GE	USD	10/4/2020	4/12/2021	4.86	4.60	3.55	3.28	B+	B+
LK	GE	USD	7/27/2021	4/12/2021	4.97	4.60	3.66	3.28	B+	B+
LK	GE	USD	10/4/2020	4/12/2021	4.88	4.60	3.57	3.28	B+	B+
LK	GE	USD	7/27/2021	4/12/2021	4.97	4.60	3.66	3.28	B+	B+
LK	JO	USD	1/22/2015	11/12/2015	3.31	4.69	3.14	4.45	B+	BB
LK	RS	USD	7/25/2022	4/9/2023	5.05	5.60	3.74	4.29	B+	BB
LK	RS	USD	7/25/2022	4/9/2023	5.04	5.60	3.73	4.29	B+	BB
MX	AE	USD	2/17/2014	4/8/2014	0.79	0.38	0.65	0.23	BBB+	A
MX	AE	USD	1/15/2014	4/8/2014	0.60	0.38	0.46	0.23	BBB+	A
MX	AE	USD	12/30/2019	4/8/2019	1.46	2.09	0.91	1.54	BBB+	A
MX	AE	USD	3/19/2019	4/8/2019	1.97	2.09	1.42	1.54	BBB+	A
MX	LT	USD	1/15/2020	11/2/2020	2.17	2.93	1.62	1.62	BBB+	BBB
MX	LT	EUR	6/16/2015	10/6/2015	0.80	0.73	0.58	0.51	BBB+	BBB
MX	LV	EUR	5/8/2017	5/3/2018	2.00	2.05	1.62	1.48	BBB+	BBB
MX	LV	EUR	7/14/2017	5/3/2018	1.81	2.05	1.24	1.48	BBB+	BBB
MX	MY	USD	1/15/2017	7/6/2016	1.43	1.63	1.06	1.26	BBB+	A–
MX	MY	USD	9/15/2016	7/6/2016	1.02	1.63	0.65	1.26	BBB+	A–
MX	MY	USD	3/15/2022	7/6/2021	2.49	2.73	1.18	1.42	BBB+	A–
MX	RU	USD	9/24/2022	4/4/2022	2.58	2.98	1.27	1.67	BBB+	BBB
MX	RU	USD	3/3/2015	4/29/2015	0.93	1.07	0.76	0.90	BBB+	BBB
MX	SK	EUR	6/10/2013	5/20/2014	0.24	0.34	0.19	0.29	BBB+	A
PA	BB	USD	1/26/2036	12/5/2035	3.91	7.50	1.75	5.34	BBB	BBB
PA	LT	USD	1/30/2020	11/2/2020	2.08	2.93	1.53	1.62	BBB	BBB
PA	LV	USD	1/16/2023	4/6/2022	3.74	3.38	2.43	2.07	BBB	BBB
PA	MY	USD	3/15/2015	6/4/2015	1.19	1.29	1.02	1.05	BBB	A–
PA	RU	USD	9/30/2027	6/26/2028	3.46	4.14	1.51	2.19	BBB	BBB

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Matched pairs: Directly treated bonds (continued)

Country	Currency	Maturity	Yield	Credit spread	Rating
PA	RU	USD	4/1/2029	6/26/2028	3.60 4.14 1.64 2.19 BBB BBB
PH	HR	EUR	3/15/2016	5/6/2015	1.81 2.68 1.60 2.58 BB BBB–
PH	HR	USD	1/20/2020	3/4/2019	2.07 3.73 1.52 3.18 BB BBB–
PH	HR	USD	1/15/2021	2/7/2021	2.22 4.21 0.90 2.90 BB BBB–
PH	HR	USD	6/17/2019	3/4/2019	2.02 3.73 1.47 3.18 BB BBB–
PH	HR	USD	1/15/2019	3/4/2019	1.95 3.73 1.40 3.18 BB BBB–
PH	JO	USD	3/17/2015	11/12/2015	0.81 4.69 0.64 4.45 BB BB
PH	JO	USD	1/15/2016	11/12/2015	1.10 4.69 0.86 4.45 BB BB
PH	RS	USD	9/25/2024	1/4/2024	2.89 6.56 1.58 5.25 BB BB
PH	RS	USD	10/21/2024	1/4/2024	2.98 6.56 1.67 5.25 BB BB
PH	UA	USD	1/15/2014	6/11/2013	0.79 5.88 0.65 5.74 BB B+
PH	UA	USD	3/16/2025	4/11/2024	3.10 7.67 1.78 6.36 BB B+
PH	UA	USD	1/18/2017	5/6/2017	1.44 6.89 1.07 6.52 BB B+
PL	AE	USD	1/15/2014	4/8/2014	0.49 0.38 0.35 0.23 A– A
PL	BB	USD	7/10/2036	12/5/2035	3.67 7.50 1.51 5.34 A– BBB
PL	LT	USD	3/7/2020	11/2/2020	2.32 2.96 1.77 1.65 A– BBB
PL	MY	USD	7/10/2016	7/6/2016	0.96 1.63 0.59 1.26 A– A–
PL	MY	USD	7/16/2016	7/6/2016	0.96 1.63 0.59 1.26 A– A–
PL	MY	USD	4/21/2021	7/6/2021	2.67 2.73 1.35 1.42 A– A–
PL	SK	USD	11/11/2023	9/5/2023	2.83 3.39 1.52 2.08 A– A
PL	SK	USD	5/9/2024	9/5/2023	3.04 3.39 1.72 2.08 A– A
QA	AT	USD	4/9/2014	5/19/2014	0.71 0.48 0.57 0.34 AA AA+
QA	AT	USD	1/20/2017	5/6/2017	1.62 0.74 1.25 0.37 AA AA+
QA	AT	USD	1/20/2017	5/6/2017	1.62 0.74 1.25 0.37 AA AA+
QA	AT	USD	1/20/2015	5/19/2014	1.15 0.48 0.98 0.34 AA AA+
QA	AT	USD	1/20/2015	5/19/2014	1.18 0.48 1.01 0.34 AA AA+
QA	AT	USD	4/9/2014	5/19/2014	0.75 0.48 0.61 0.34 AA AA+
RO	HR	USD	7/2/2022	12/3/2022	4.44 4.29 3.13 2.98 BB+ BBB–
SV	RS	USD	1/24/2023	4/9/2023	5.47 5.60 4.16 4.29 BB– BB
SV	UA	USD	12/1/2019	12/7/2018	4.76 7.25 4.21 6.70 BB– B+
TR	HR	USD	3/25/2022	12/3/2022	3.21 4.28 1.90 2.97 BB BBB–
TR	HR	USD	3/11/2019	3/4/2019	2.69 3.73 2.14 3.18 BB BBB–
TR	HR	USD	6/5/2020	2/7/2021	2.92 4.21 1.61 2.90 BB BBB–
TR	HR	USD	3/30/2021	2/7/2021	3.07 4.21 1.76 2.90 BB BBB–
TR	HR	USD	11/7/2019	3/4/2019	2.84 3.73 2.29 3.18 BB BBB–
TR	HR	USD	9/26/2022	12/3/2022	3.27 4.28 1.95 2.97 BB BBB–
TR	HR	USD	4/3/2018	3/4/2019	2.54 3.73 1.99 3.18 BB BBB–
TR	JO	USD	9/26/2016	11/12/2015	2.29 4.69 1.91 4.45 BB BB
TR	JO	USD	3/15/2015	11/12/2015	1.89 4.69 1.72 4.45 BB BB
TR	UA	USD	7/14/2017	5/6/2017	2.42 6.89 1.87 6.52 BB B+

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Matched pairs: Directly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
TR	UA	USD	2/5/2025	4/11/2024	3.70	7.67	2.39	6.36	BB	B+
TR	UA	USD	1/15/2014	6/11/2013	1.37	5.88	1.22	5.74	BB	B+
ZA	AE	USD	6/23/2017	5/2/2017	3.34	3.42	2.78	3.05	BBB+	A
ZA	LT	USD	5/30/2022	9/3/2021	3.12	3.13	1.80	1.82	BBB+	BBB
ZA	LT	EUR	4/5/2016	2/10/2016	1.76	1.55	1.55	1.34	BBB+	BBB
ZA	RU	USD	5/27/2019	4/29/2020	2.56	2.57	2.01	2.02	BBB+	BBB
ZA	RU	USD	3/9/2020	4/29/2020	2.76	2.60	2.21	2.05	BBB+	BBB
ZA	RU	USD	6/2/2014	4/29/2015	0.92	1.07	0.75	0.90	BBB+	BBB
ZA	SK	USD	1/17/2024	9/5/2023	3.33	3.39	2.02	2.08	BBB+	A

Table SA.5: **Matched pairs, New-York law experiment: Indirectly treated bonds.** We list 110 indirectly treated bond pairs below with bond characteristics for treated (local-law/New York-law issuer, left) and control (local-law/New York-law non-issuer, right) bonds in each column. Yields and credit spreads are as of the day of the first event (November 26, 2012).

Country		Currency	Maturity		Yield		Credit spread		Rating	
BG	LT	EUR	3/30/2016	10/20/2016	1.93	2.09	1.71	1.72	BBB	BBB
BG	LV	EUR	1/18/2019	8/10/2018	2.60	2.41	2.03	1.83	BBB	BBB
BG	LV	EUR	10/11/2018	8/10/2018	2.46	2.41	1.89	1.83	BBB	BBB
FI	DE	EUR	3/3/2015	4/10/2015	0.02	0.02	-0.09	-0.09	AAA	AAA
FI	DE	EUR	7/4/2013	7/4/2013	-0.01	-0.01	-0.06	-0.06	AAA	AAA
FI	DE	EUR	7/4/2015	10/9/2014	0.11	-0.00	-0.10	-0.10	AAA	AAA
FI	FR	EUR	4/10/2025	10/25/2025	2.03	2.51	0.62	0.63	AAA	AAA
FI	FR	EUR	4/15/2016	7/12/2015	0.26	0.26	0.05	0.05	AAA	AAA
FI	FR	EUR	4/4/2019	4/25/2018	1.07	1.05	0.50	0.48	AAA	AAA
FI	LU	EUR	4/15/2020	5/18/2020	1.25	1.29	0.68	0.72	AAA	AAA
FI	NL	EUR	3/5/2018	3/7/2018	0.61	0.61	0.03	0.04	AAA	AAA
FI	NL	EUR	4/15/2017	1/15/2017	0.49	0.51	0.12	0.13	AAA	AAA
FI	NL	EUR	3/9/2023	7/15/2022	1.68	1.69	0.27	0.28	AAA	AAA
FI	NL	EUR	4/15/2021	7/15/2021	1.46	1.51	0.05	0.10	AAA	AAA
FI	NL	EUR	7/4/2028	1/15/2028	2.23	2.23	0.36	0.35	AAA	AAA
FI	NL	EUR	7/4/2042	1/15/2042	2.49	2.48	0.38	0.38	AAA	AAA
IT	ES	EUR	9/1/2022	1/31/2022	4.73	5.64	3.32	4.23	BBB+	BBB+
IT	ES	EUR	2/1/2015	11/2/2014	2.15	2.24	2.05	2.14	BBB+	BBB+
IT	ES	EUR	9/15/2041	7/30/2041	3.71	6.32	1.60	4.21	BBB+	BBB+
IT	ES	EUR	2/10/2014	6/1/2014	0.31	0.95	0.26	0.85	BBB+	BBB+
IT	ES	EUR	4/1/2014	5/5/2014	1.72	1.80	1.67	1.76	BBB+	BBB+
IT	ES	EUR	11/1/2026	7/30/2026	5.19	6.09	3.31	4.21	BBB+	BBB+

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Matched pairs: Indirectly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
IT	ES	EUR	9/15/2023	1/31/2024	3.29	5.89	1.88	4.47	BBB+	BBB+
IT	ES	EUR	1/5/2015	10/11/2014	2.55	2.59	2.45	2.49	BBB+	BBB+
IT	ES	EUR	11/1/2029	1/31/2029	5.28	6.11	3.40	4.23	BBB+	BBB+
IT	ES	EUR	12/15/2013	3/3/2014	1.54	1.54	1.49	1.49	BBB+	BBB+
IT	ES	EUR	6/5/2016	6/10/2015	1.93	1.65	1.55	1.43	BBB+	BBB+
IT	ES	EUR	4/8/2015	6/1/2014	0.51	0.95	0.41	0.85	BBB+	BBB+
IT	ES	EUR	2/1/2037	1/31/2037	5.32	6.31	3.28	4.28	BBB+	BBB+
IT	ES	EUR	3/1/2022	4/30/2021	4.59	5.53	3.18	4.12	BBB+	BBB+
IT	ES	EUR	6/1/2017	6/3/2017	3.50	3.33	2.93	2.76	BBB+	BBB+
IT	ES	EUR	4/15/2016	9/8/2015	2.80	2.76	2.58	2.54	BBB+	BBB+
IT	ES	EUR	1/1/2017	6/3/2017	3.22	3.33	2.84	2.76	BBB+	BBB+
IT	ES	EUR	11/1/2013	3/8/2014	1.41	1.36	1.36	1.31	BBB+	BBB+
IT	ES	EUR	12/22/2023	1/31/2024	4.88	5.89	3.47	4.47	BBB+	BBB+
IT	ES	EUR	2/3/2014	6/1/2014	0.55	0.95	0.50	0.85	BBB+	BBB+
IT	ES	EUR	9/15/2014	6/1/2014	0.62	0.95	0.52	0.85	BBB+	BBB+
IT	ES	EUR	5/4/2027	7/30/2026	5.63	6.09	3.76	4.21	BBB+	BBB+
IT	ES	EUR	4/15/2018	6/3/2017	3.24	3.33	2.67	2.76	BBB+	BBB+
IT	ES	EUR	12/7/2013	3/8/2014	1.17	1.36	1.12	1.31	BBB+	BBB+
IT	ES	EUR	11/1/2022	1/31/2022	4.78	5.64	3.37	4.23	BBB+	BBB+
IT	ES	EUR	8/1/2023	1/31/2024	4.73	5.89	3.32	4.47	BBB+	BBB+
IT	ES	EUR	8/1/2014	9/12/2014	1.91	1.93	1.81	1.83	BBB+	BBB+
IT	ES	EUR	7/25/2016	7/30/2015	3.45	3.45	3.07	3.23	BBB+	BBB+
IT	ES	EUR	2/1/2014	6/1/2014	0.29	0.95	0.24	0.85	BBB+	BBB+
IT	ES	EUR	3/1/2015	11/2/2014	2.27	2.24	2.17	2.14	BBB+	BBB+
IT	ES	EUR	8/1/2017	6/3/2017	3.50	3.33	2.93	2.76	BBB+	BBB+
IT	ES	EUR	2/8/2014	3/8/2014	1.23	1.36	1.18	1.31	BBB+	BBB+
IT	ES	EUR	1/31/2014	7/10/2014	1.66	1.69	1.61	1.59	BBB+	BBB+
IT	ES	EUR	5/1/2017	6/3/2017	3.44	3.33	3.07	2.76	BBB+	BBB+
IT	ES	EUR	7/1/2014	5/5/2014	1.85	1.80	1.75	1.76	BBB+	BBB+
IT	ES	EUR	7/15/2015	8/9/2014	2.52	2.37	2.31	2.27	BBB+	BBB+
IT	ES	EUR	1/9/2014	3/8/2014	1.29	1.36	1.25	1.31	BBB+	BBB+
IT	ES	EUR	1/1/2022	4/30/2021	3.37	5.53	1.96	4.12	BBB+	BBB+
IT	ES	EUR	3/1/2014	6/1/2014	0.17	0.95	0.12	0.85	BBB+	BBB+
IT	ES	EUR	9/1/2040	7/30/2041	5.49	6.32	3.38	4.21	BBB+	BBB+
IT	ES	EUR	12/4/2013	6/1/2014	0.79	0.95	0.74	0.85	BBB+	BBB+
IT	ES	EUR	6/1/2014	5/5/2014	1.84	1.80	1.74	1.76	BBB+	BBB+
IT	ES	EUR	9/1/2019	6/2/2020	3.99	4.74	3.42	3.33	BBB+	BBB+
IT	ES	EUR	11/12/2022	1/31/2022	4.89	5.64	3.48	4.23	BBB+	BBB+
IT	ES	EUR	9/30/2014	7/7/2014	2.09	2.02	1.99	1.92	BBB+	BBB+
IT	ES	EUR	2/10/2014	3/8/2014	1.36	1.36	1.32	1.31	BBB+	BBB+

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Matched pairs: Indirectly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
IT	ES	EUR	7/11/2014	6/1/2014	0.74	0.95	0.65	0.85	BBB+	BBB+
IT	ES	EUR	2/1/2024	1/31/2024	4.54	5.89	3.13	4.47	BBB+	BBB+
IT	ES	EUR	4/9/2015	6/1/2014	0.65	0.95	0.55	0.85	BBB+	BBB+
IT	ES	EUR	11/1/2023	1/31/2024	5.00	5.89	3.59	4.47	BBB+	BBB+
IT	ES	EUR	9/15/2016	6/3/2017	3.04	3.33	2.66	2.76	BBB+	BBB+
IT	ES	EUR	2/11/2014	3/8/2014	1.42	1.36	1.37	1.31	BBB+	BBB+
IT	ES	EUR	8/1/2013	3/8/2014	1.30	1.36	1.25	1.31	BBB+	BBB+
IT	ES	EUR	2/2/2014	6/1/2014	0.45	0.95	0.40	0.85	BBB+	BBB+
IT	ES	EUR	5/5/2015	5/5/2014	1.82	1.80	1.72	1.76	BBB+	BBB+
IT	ES	EUR	8/2/2013	6/1/2014	0.15	0.95	0.10	0.85	BBB+	BBB+
IT	ES	EUR	8/1/2016	9/8/2015	2.96	2.76	2.59	2.54	BBB+	BBB+
IT	ES	EUR	11/15/2014	7/7/2014	2.03	2.02	1.92	1.92	BBB+	BBB+
IT	IE	EUR	9/15/2016	6/2/2016	1.91	1.43	1.53	1.06	BBB+	BBB+
IT	IE	EUR	9/15/2021	10/18/2020	3.06	4.40	1.65	2.99	BBB+	BBB+
IT	IE	EUR	3/1/2016	4/18/2016	2.37	2.38	2.16	2.16	BBB+	BBB+
IT	IE	EUR	3/1/2025	3/13/2025	4.97	4.95	3.56	3.54	BBB+	BBB+
IT	IE	EUR	2/3/2018	10/18/2017	2.07	3.11	1.50	2.54	BBB+	BBB+
IT	IE	EUR	2/1/2033	8/9/2033	5.53	5.53	3.50	3.50	BBB+	BBB+
IT	IE	EUR	1/2/2044	8/9/2043	3.84	5.90	1.73	3.79	BBB+	BBB+
IT	IE	EUR	11/1/2017	10/18/2018	3.62	3.81	3.05	3.23	BBB+	BBB+
IT	IE	EUR	6/16/2015	4/18/2016	2.42	2.38	2.20	2.16	BBB+	BBB+
IT	IE	EUR	6/9/2015	6/2/2016	1.37	1.43	1.15	1.06	BBB+	BBB+
IT	IE	EUR	2/1/2020	6/18/2019	4.12	4.13	3.55	3.56	BBB+	BBB+
IT	IE	EUR	8/1/2034	8/9/2033	5.44	5.53	3.41	3.50	BBB+	BBB+
IT	IE	EUR	11/1/2015	4/18/2016	2.56	2.38	2.35	2.16	BBB+	BBB+
IT	IE	EUR	3/1/2019	10/18/2018	3.94	3.81	3.37	3.23	BBB+	BBB+
IT	IE	EUR	3/17/2021	6/6/2020	3.53	4.19	2.12	2.78	BBB+	BBB+
IT	IE	EUR	8/1/2018	10/18/2018	3.78	3.81	3.21	3.23	BBB+	BBB+
IT	IE	EUR	3/1/2026	3/13/2025	5.00	4.95	3.12	3.54	BBB+	BBB+
IT	IE	EUR	9/15/2017	10/18/2017	2.23	3.11	1.66	2.54	BBB+	BBB+
IT	IE	EUR	2/1/2018	10/18/2018	3.65	3.81	3.08	3.23	BBB+	BBB+
IT	IE	EUR	8/1/2021	10/18/2020	4.41	4.40	3.00	2.99	BBB+	BBB+
IT	IE	EUR	8/1/2039	8/9/2038	5.49	5.75	3.38	3.64	BBB+	BBB+
IT	IE	EUR	1/4/2020	6/18/2019	4.17	4.13	3.60	3.56	BBB+	BBB+
IT	IE	EUR	9/1/2021	10/18/2020	4.44	4.40	3.03	2.99	BBB+	BBB+
IT	IE	EUR	5/3/2028	8/9/2028	3.42	5.12	1.54	3.24	BBB+	BBB+
IT	IE	EUR	6/15/2017	10/18/2017	3.15	3.11	2.58	2.54	BBB+	BBB+
IT	IE	EUR	11/1/2027	8/9/2028	5.33	5.12	3.45	3.24	BBB+	BBB+
IT	IE	EUR	10/10/2017	10/18/2017	2.13	3.11	1.56	2.54	BBB+	BBB+
IT	IE	EUR	3/1/2020	6/18/2019	4.11	4.13	3.54	3.56	BBB+	BBB+

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Matched pairs: Indirectly treated bonds (continued)

Country		Currency	Maturity		Yield		Credit spread		Rating	
IT	IE	EUR	3/1/2021	10/18/2020	4.30	4.40	2.89	2.99	BBB+	BBB+
IT	IE	EUR	7/7/2015	6/2/2016	0.41	1.43	0.20	1.06	BBB+	BBB+
IT	IE	EUR	11/6/2016	4/18/2016	2.24	2.38	1.87	2.16	BBB+	BBB+
IT	IE	EUR	6/10/2015	6/2/2016	0.86	1.43	0.65	1.06	BBB+	BBB+
IT	IE	EUR	10/15/2017	10/18/2017	3.14	3.11	2.57	2.54	BBB+	BBB+
IT	IE	EUR	12/15/2015	4/18/2016	2.48	2.38	2.26	2.16	BBB+	BBB+
IT	IE	EUR	9/15/2019	6/6/2020	2.63	4.19	2.06	2.78	BBB+	BBB+
IT	IE	EUR	2/1/2019	10/18/2018	3.89	3.81	3.31	3.23	BBB+	BBB+

Table SA.6: **Summary statistics on the event date, by country.** We report statistics for the 67 countries with bonds in our final sample on the event date, Thursday, May 31, 2012, following the announcement of the ruling on May 30.

Country	Bonds	Yield	Credit spread	Maturity	Total (\$bn)	English law (%)	NY law (%)	Local law (%)
AE	13	3.73	2.62	9.84	15.56	1.00	–	–
AT	33	1.69	0.81	10.02	91.41	0.58	–	0.37
BB	4	6.92	4.30	14.42	1.20	0.90	–	0.10
BE	51	1.90	1.15	10.88	193.89	> 0	–	< 1
BG	23	3.28	2.85	4.97	6.84	0.17	0.56	0.24
BH	6	4.53	3.38	7.06	11.24	0.99	–	–
BR	35	5.32	1.86	8.66	1284.43	> 0	0.03	0.97
BS	6	6.03	4.06	18.99	1.80	–	1.00	–
BY	2	10.99	10.57	4.42	1.80	1.00	–	–
CI	2	10.37	8.38	20.60	5.22	–	1.00	–
CL	12	2.86	-1.43	14.68	13.15	–	0.38	0.62
CO	23	4.25	1.04	11.25	115.29	–	0.14	0.84
CR	4	4.58	3.32	12.67	46.68	–	0.16	0.68
CY	3	13.85	11.51	4.47	5.11	0.76	–	–
CZ	27	2.25	1.02	9.14	20.50	0.71	–	0.29
DE	66	0.51	-0.22	7.49	509.41	–	–	0.99
DK	17	0.61	-0.41	6.50	40.09	0.55	–	0.45
DO	7	6.31	4.05	8.40	13.38	–	0.80	0.20
EC	2	13.85	11.51	23.20	10.02	0.20	0.80	–
EG	3	7.33	4.72	12.94	26.55	0.14	0.16	0.70
ES	53	4.75	4.11	6.79	459.65	0.03	–	0.97
FI	23	1.12	0.20	7.04	80.02	0.26	> 0	0.74
FR	64	1.22	0.42	9.45	387.58	–	–	1.00
GE	2	5.89	3.05	8.87	1.00	1.00	–	–
GH	2	5.95	5.37	7.00	2.96	0.93	–	–
GR	20	13.85	11.51	17.61	360.71	0.46	–	0.53
GT	4	5.16	3.41	12.32	4.06	–	1.00	–
HR	22	6.11	5.13	7.05	26.76	0.77	–	0.23
HU	34	8.01	4.55	6.39	32.93	0.39	0.48	0.08
ID	56	4.77	2.64	8.16	213.17	0.04	0.28	0.67
IE	10	6.92	6.15	17.03	71.34	> 0	–	< 1
IL	7	2.44	1.63	8.88	13.95	0.43	0.57	–
IM	1	3.37	0.69	20.03	0.40	1.00	–	–
IS	2	5.44	3.73	12.77	4.90	0.24	0.20	0.55
IT	124	4.48	3.73	8.04	968.05	> 0	0.02	0.98
JM	8	7.53	6.23	10.85	4.18	–	0.87	0.08
JO	1	4.36	4.11	3.45	0.75	1.00	–	–

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Summary statistics by country (continued)

Country	Bonds	Yield	Credit spread	Maturity	Total (\$bn)	English law (%)	NY law (%)	Local law (%)
LB	21	4.92	4.03	6.64	21.75	0.06	0.87	0.08
LK	5	6.18	3.86	7.72	11.90	–	0.80	0.20
LT	15	4.35	2.93	5.55	20.09	0.98	–	0.02
LU	3	1.17	0.27	12.91	9.95	–	–	1.00
LV	5	4.38	3.41	6.06	9.94	0.98	–	0.02
ME	2	9.05	8.60	5.71	1.26	1.00	–	–
MK	2	5.50	5.22	2.19	0.50	0.81	–	0.19
MX	39	3.64	1.53	12.29	282.26	–	0.16	0.81
MY	45	3.26	0.23	8.20	208.83	0.04	–	0.96
NG	9	12.94	2.85	8.10	4.95	0.61	–	0.39
NL	28	0.88	0.18	7.28	306.46	–	–	1.00
PA	12	3.65	2.00	13.46	10.86	–	0.48	0.47
PE	12	3.89	1.99	14.33	21.03	–	0.35	0.65
PH	65	4.96	1.05	12.62	88.05	0.01	0.26	0.71
PL	62	3.70	1.22	8.71	87.85	0.50	0.14	0.29
PT	21	8.30	7.40	4.89	83.04	0.06	–	0.94
QA	18	3.43	1.93	11.43	38.05	0.14	0.86	–
RO	21	5.71	2.04	7.67	26.71	0.68	0.06	0.26
RS	2	7.28	4.44	8.83	12.65	1.00	–	–
RU	34	6.29	0.72	9.13	82.00	0.83	–	0.17
SE	22	0.78	-0.86	7.03	204.61	0.17	> 0	0.82
SI	10	4.58	3.71	8.04	37.04	0.57	–	0.43
SK	25	2.44	1.64	7.79	19.35	0.50	–	0.42
SV	7	6.82	4.57	17.19	9.69	–	< 1	> 0
TR	50	5.93	1.43	7.71	129.75	0.09	0.29	0.58
TT	3	4.44	2.29	12.26	2.04	0.07	0.93	–
UA	15	10.00	9.08	5.74	40.68	0.81	–	0.18
UY	15	3.63	2.15	13.64	15.92	–	0.83	0.12
VE	14	12.07	10.46	9.93	48.45	–	0.71	0.29
ZA	24	4.42	0.00	13.54	203.47	–	0.06	0.94